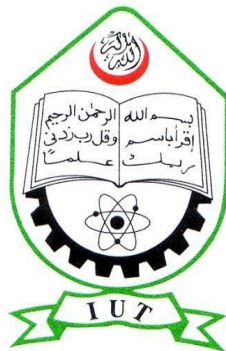


PROJECT TITLE
**Machine Learning-Driven Handover
Optimization in LTE networks: Quality Signal Indicator,
Classifier-Based Prediction Analysis and Hybrid
Q-Learning.**

by

**K.M. Istiaque. (ID: 200021220)
M.M. Sadman Shafi (ID: 200021240)
Shafayet Sadik Sowad (ID: 200021248)**

**Report on
EEE 4700 (Project and Thesis)**



Department of Electrical and Electronic Engineering
Islamic University of Technology (IUT)
Gazipur, Bangladesh

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THESIS TITLE
Machine Learning-Driven Handover
Optimization in LTE networks: Quality Signal
Indicator, Classifier-Based Prediction Analysis and
Hybrid Q-Learning.

Approved by:

Dr. Mohammad Tawhid Kawser

Supervisor and Professor,
Department of Electrical and Electronic Engineering,
Islamic University of Technology (IUT),
Boardbazar, Gazipur-1704.

Date:

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List of Acronyms

Acronym	Full Form
AM	Acknowledged Mode
APN	Access Point Name
CINR	Carrier-to-Interference-plus-Noise Ratio
CO	Course Outcome
CQI	Channel Quality Indicator
DQN	Deep Q-Network
DRM	Digital Rights Management
EPC	Evolved Packet Core
E-UTRAN	Evolved Universal Terrestrial Radio Access Network
EPS	Evolved Packet System
GPS	Global Positioning System
KPI	Key Performance Indicator
LTE	Long Term Evolution
MME	Mobility Management Entity
ML	Machine Learning
MSV	Multiple Site Verification
MNO	Mobile Network Operator
PCA	Principal Component Analysis

PGW	Packet Data Network Gateway
Q-Learning	Quality-Learning
QoS	Quality of Service
RLC	Radio Link Control
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RL	Reinforcement Learning
RSSI	Received Signal Strength Indicator
SAE	System Architecture Evolution
SVM	Support Vector Machine
SNR	Signal-to-Noise Ratio
SGW	Serving Gateway
SSV	Single Site Verification
TTT	Time to Trigger
UE	User Equipment
XCAL-M	Professional Drive Test Software (Accuver Product)
3GPP	3rd Generation Partnership Project

Chapter 1

Introduction

1.1 Overview of LTE network

Long Term Evolution (LTE) is a standardized fourth-generation (4G) wireless communication system developed by the 3rd Generation Partnership Project (3GPP), designed to provide high-speed data rates, low latency, and robust support for mobility. LTE is a fully packet-switched architecture, which eliminates the need for circuit-switched elements used in earlier generations. Its core network is the Evolved Packet Core (EPC), which integrates components such as the Mobility Management Entity (MME), Serving Gateway (SGW), and Packet Data Network Gateway (PGW). The radio access network is composed of evolved NodeBs (eNodeBs), which handle both control and user-plane functionalities, and directly interface with User Equipment (UE), forming a flat, all-IP based architecture that reduces signaling overhead and latency.

The system design adheres to the System Architecture Evolution (SAE) framework, which supports scalable deployment, IP-based services, and interworking with both legacy and non-3GPP access networks. Together, the Evolved Universal Terrestrial Radio Access Network (E-UTRAN) and the EPC constitute the Evolved Packet System (EPS). EPS is optimized for high mobility and Quality of Service (QoS) provisioning across heterogeneous environments, making it suitable for urban, dense, and high-speed scenarios.

A critical aspect of LTE network performance is its ability to maintain service continuity during user mobility. This is achieved through advanced mobility management procedures, primarily handover mechanisms—specifically, intra-LTE handovers facilitated by X2 or S1 interfaces, depending on network configuration and inter-eNodeB connectivity. Handover decisions are driven by measurements such as Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), Received Signal Strength Indicator (RSSI), and Channel Quality Indicator (CQI), which the UE periodically reports to the serving eNodeB.

1.2 Overview of Drive Test

To empirically evaluate such mobility procedures, drive tests are employed to collect real-time radio access measurements under actual deployment conditions. These tests can be classified into:

- Single Site Verification (SSV): Focused on the performance validation of an individual cell,
- Multiple Site Verification (MSV): Involving coordinated testing over a group of cells within a cluster to assess inter-cell interactions,
- Operator Benchmarking: Comparing performance across different mobile network operators.

The present study undertook an MSV drive test within an LTE cellular cluster in an urban environment in Dhaka, covering a route approximately 13 kilometers in length. The urban morphology was characterized by high building density, multipath propagation, and high user

mobility—all contributing to frequent inter-cell handovers. The drive test aimed to collect a high-fidelity, mobility-aware dataset, capturing dynamic radio conditions and corresponding handover events.

Such datasets are pivotal for machine learning (ML)-based handover prediction and optimization, where models can learn from real-world handover behaviors and signal dynamics to reduce unnecessary handovers (e.g., ping-pong effects), avoid dropped calls, and improve overall user experience. This aligns with LTE's architectural goals of efficient radio resource utilization, minimal handover latency, and service continuity, particularly in dense urban deployments where user mobility and cell overlap are significant challenges.

1.3 Background of our Study

The increasing demand for seamless mobile connectivity, particularly in high-mobility environments such as highways and railways reflect the importance of optimizing handover mechanisms in wireless communication networks. Handover, which is the process of transferring a user's connection from one base station to another. It is crucial for maintaining continuous service and real world relevance in ensuring quality user experience. Traditional handover strategies based on fixed signal strength thresholds often result in unnecessary transitions or delayed decisions, negatively affecting network performance.

Therefore, this project is needed to explore modern, intelligent alternatives that can enhance handover efficiency and reliability. The core problem lies in the inefficiency of traditional handover approaches, which do not adapt well to dynamic network conditions and can lead to increased latency and frequent disconnections.

There is an existing gap in leveraging real-time data and predictive models for proactive handover decisions. Most existing methods overlook the potential of advanced data analytics and machine learning. Additionally, a major challenge encountered was the lack of publicly available datasets relevant to this study. To overcome this, a drive test was conducted to collect real-world data, which then required extensive pre-processing and cleaning before it could be used for model development.

The key goals of this study are: To utilize machine learning (ML) techniques for improving the handover decision-making process. To analyze wireless signal quality metrics such as RSRP, RSRQ, and neighboring cell conditions. To design a data-driven framework capable of predicting and optimizing handovers. This study focuses on the analysis and development of a machine learning-based handover optimization model using real-world drive test data. The scope includes wireless signal parameter analysis, data pre-processing, feature selection, and ML model training and evaluation.

Limitations include the scarcity of publicly available datasets, which necessitated the time-intensive process of data collection via drive testing. Moreover, computational constraints and the complexity of real-world signal variability posed additional challenges during data processing and model development.

Chapter 2

Literature Review

2.1 Introduction

This chapter presents a comprehensive overview of previous research and existing work related to LTE handover mechanisms, Quality of Service (QoS) evaluation, and machine learning-based optimization techniques. The goal is to contextualize the current project, which involves the collection and processing of a real-world LTE dataset from Bangladesh to develop predictive and optimization models for handover events. Existing studies in the domains of drive testing, QoS metrics, and intelligent handover management are critically examined, followed by a detailed comparison to highlight the novelty and significance of the present research.

2.2 Existing Studies

2.2.1 Drive Test-Based Datasets for LTE Analysis

Several prior studies have attempted to analyze LTE performance through field measurements and drive test data. Musa et al. [1] conducted LTE signal quality analysis in Nigeria's Sabon-Gari Market, collecting data over three weeks using drive test applications. Although useful for urban signal behavior analysis, the dataset lacked detailed handover event structures and preprocessing for ML applications.

Similarly, Schwarz et al. [2] collected LTE drive test data on an Austrian highway to support RLC throughput prediction. While this dataset was tailored for machine learning, it omitted mobility context such as UE speed and raw RRC measurements, which are critical for handover analysis.

Kpanake et al. [3] compared LTE network quality across North-Central Nigerian cities, capturing only short-term metrics without timestamped handover data or machine learning-ready formatting.

Islam et al. [4] explored RF measurements through a 2G drive test for QoS assessment, but the data lacked modern LTE-specific parameters and mobility event logs.

2.2.2 QoS-Aware Handover Techniques

Traditional LTE handover mechanisms depend on fixed thresholds of KPIs like RSRP, RSRQ, and SINR [5]. These static techniques perform well in stationary or predictable environments but often result in suboptimal performance under dynamic mobility, leading to issues like ping-pong handovers or late handovers that degrade QoS [6].

2.2.3 Machine Learning-Based Handover Prediction

To improve mobility management, ML models such as Decision Trees, SVMs, and Neural Networks have been proposed to predict handover triggers based on historical data [3], [7]. Deep learning approaches using CNNs and RNNs offer higher accuracy by modeling spatial and temporal features simultaneously, but they demand high computational resources and are sensitive to data quality [4], [8].

2.2.4 Reinforcement Learning for Handover Optimization

Reinforcement learning (RL) algorithms like Q-learning have gained popularity due to their adaptive nature. These methods optimize handover policies by learning from environmental feedback, thus reducing unnecessary handovers [5], [6]. Recent advances, such as Deep Q-Networks (DQNs), further improve policy efficiency in high-dimensional state spaces [7]. However, their deployment in real-world networks is constrained by the lack of training data and simulation-heavy methodologies [9], [10].

2.3 Comparison with Existing Work

This research builds upon existing methods by integrating both ML and RL techniques with real-world LTE measurements collected through drive tests in Bangladesh. Unlike prior studies that rely on simulations or limited mobility contexts, this study includes timestamped handover events, parsed measurement reports, dynamic UE velocities, and contextual network parameters.

A key contribution of this project is the feature engineering process, which incorporates Time to Trigger (TTT), noise reduction, and domain-specific selection to improve model interpretability and efficiency [12]. The inclusion of DRM states, raw and processed data stages, and coverage context makes the dataset uniquely robust for ML-based handover prediction and QoS optimization.

Unlike the limited or simulation-driven datasets used in previous ML/RL studies [9], [14], this dataset is structured specifically for real-world validation of mobility prediction models. Moreover, its publication fills a geographic and demographic research gap, being the first LTE handover dataset publicly shared from Bangladesh.

2.4 Summary

In summary, existing literature highlights the growing interest in ML and RL techniques for mobility optimization in LTE networks. However, significant gaps remain in terms of real-world validation, data availability, and feature completeness. This study addresses these gaps by offering a regionally specific, ML-ready dataset from live drive tests and by

integrating modern learning techniques tailored for real-world application in handover prediction and QoS optimization.

Chapter 3

Drive Test

This section encompasses details about the drive test location, time period, required equipment, software set up and data extraction methods.

3.1 Equipment Used in the Drive Test:

- XCAL-M software
- Samsung device with Exynos SoC chipset
- Laptop
- Android-based mobile phone
- Arduino Uno
- Neo-7M GPS module
- Vehicle

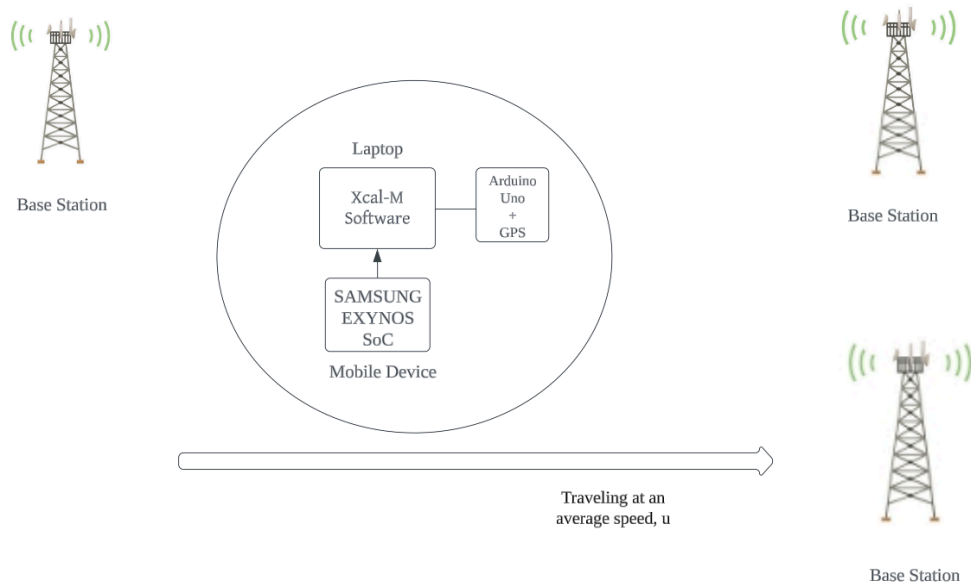


Figure1: Drive Test Setup

3.2 Location of the Drive Test:

To select an appropriate location for the drive test, we utilized the publicly available cell ID database from the OpenCellID website, which provides approximate geolocations of cellular base stations. This database includes information for multiple Radio Access Technologies (RATs), namely GSM, UMTS, and LTE. For the purpose of this study, which focuses on LTE handover behavior, we specifically filtered and targeted LTE eNodeBs. We selected a region within Dhaka, Bangladesh characterized by a high density of LTE base stations to ensure frequent inter-cell handover events, which are critical for evaluating handover decision algorithms and mobility management performance.

The drive test route spanned approximately 13 kilometers, beginning at Le Meridien Dhaka (Uttara) and concluding at the BRAC University campus. This route includes variations in user mobility patterns such as stationary periods, pedestrian speeds, and vehicular movement under mixed traffic conditions. The measurements were collected using the Grameenphone LTE network, one of the largest mobile network operators in Bangladesh.

It is important to note that, due to the specific selection of the test area the use of a single Mobile Network Operator (MNO), the results obtained from this drive test may not be fully representative of broader network conditions across the city. Additionally, the findings may not generalize to other network operators with differing infrastructure deployments, frequency bands, or mobility management strategies.

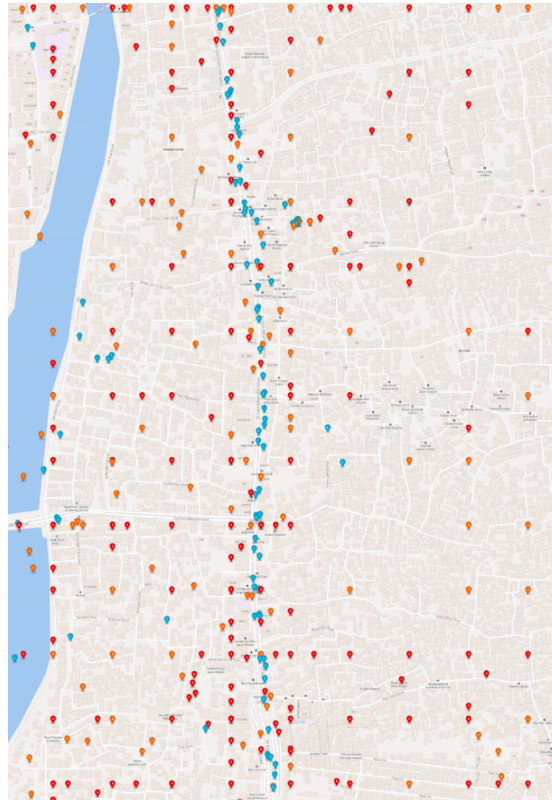


Figure 2: Open Cell ID website view of drive test location.

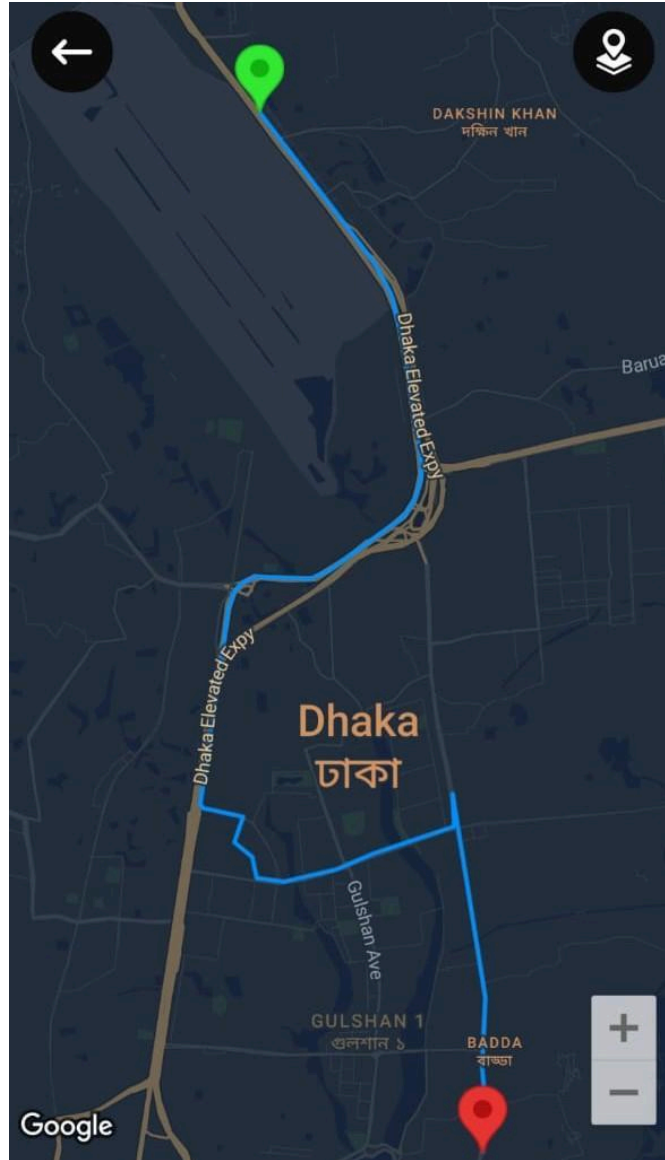


Figure 3: Drive Test Real Map.

3.3 Time Period

The drive test was conducted over three separate days: October 15, November 6, and November 16, with measurements recorded during the evening hours between 16:00 and 22:00 (local time). On each day, the test route—spanning from Le Meridien Dhaka (Uttara) to BRAC University and back—was traversed in two complete rounds, with a 30-minute interval between each outbound and return trip. The time window was deliberately chosen to coincide with evening peak hours, a period characterized by high user density and network load, which can significantly influence radio resource management and handover dynamics. The selected route lies in one of the busiest corridors of Dhaka city, commonly subject to vehicular congestion and traffic delays. Such conditions introduced non-uniform mobility patterns (e.g., stop-and-go movement), which in turn affect radio measurements, particularly

those related to handover triggering, RSRP fluctuation, and SINR degradation due to prolonged exposure to edge-of-cell conditions.

Traffic-induced delays allowed for denser spatial sampling of radio conditions, especially near cell edges. The short, repeatedly traversed route led to multiple encounters with the same LTE cell IDs under varying conditions, enhancing the dataset’s temporal diversity. This redundancy improves the robustness and generalizability of machine learning models for handover prediction.

The network infrastructure in the area remained operationally static during the measurement period; that is, no observable commissioning or decommissioning of LTE eNodeBs took place. Therefore, temporal variations in collected Key Performance Indicators (KPIs) are attributed primarily to changes in user mobility and traffic conditions, rather than alterations in network topology or configuration. The average ambient temperature and relative humidity during the test periods were approximately 29°C and 72%, respectively—values typical of Dhaka's tropical climate in the post-monsoon season.

3.4 Software Set Up

For the execution of the drive test, we employed XCAL-M, a professional-grade drive test software developed by Accuver, installed on a laptop and interfaced with a Samsung Galaxy S10 device powered by the Exynos chipset. This setup enabled the collection of detailed Layer 1 to Layer 3 signaling information, radio frequency (RF) metrics, and protocol-level data essential for in-depth analysis of mobility and handover performance in LTE networks. The mobile device was connected via diagnostic mode to ensure direct access to chipset-level KPIs.

In addition, we integrated an Arduino Uno microcontroller with a Neo-7M GPS module to continuously log the user equipment’s geolocation (Latitude and Longitude) and UE velocity. This auxiliary setup ensured accurate tracking of mobility patterns, enabling correlation between user movement and observed network behavior.

Compared to Android-based applications such as G-NetTrack Pro, which typically record measurements at one-second intervals due to OS-level constraints, XCAL-M offers millisecond-level granularity, making it substantially more suitable for fine-grained temporal and spatial analysis of handover events and transient radio conditions.

The Samsung S10 device, serving as the user equipment (UE), was placed inside a private test vehicle, and the entire system was configured for real-time data acquisition as the vehicle traversed the defined route. Throughout the drive, XCAL-M continuously captured network performance metrics and handover signaling messages, synchronized with precise GPS coordinates and time stamps.

3.5 Data Extraction from XCAL-M

Following the completion of the drive test, network performance data was extracted using the built-in post-processing capabilities of XCAL-M. To analyze mobility and handover-related events, we navigated to the “Layer3 KPI” module, selected “LTE”, and then accessed the “LTE Handover Event Statistics” window. This interface provided detailed handover metrics, including the number of handover attempts, successful handovers, and corresponding event

types along with precise timestamps, enabling temporal correlation with user movement and signal conditions.

Time	Event	Freq	Base	Result	Attempt	Success
17:32:30.189	Intra LTE-HO	(-/-) to (40490/365)	X2	Attempt	1	0
17:32:30.189	Intra LTE-HO	(-/-) to (40490/365)	X2	Success	0	1
17:32:45.091	Idle Mode Load Balancing	EUTRA(40292/7) EUTRA(40688/7)		Attempt	1	0
17:33:09.172	Idle Mode Load Balancing	EUTRA(40688/7) EUTRA(40490/365)		Attempt	1	0
17:34:31.196	Idle Mode Load Balancing	EUTRA(40688/7) EUTRA(40292/7)		Attempt	1	0
17:35:13.078	Idle Mode Load Balancing	EUTRA(275/7) EUTRA(1600/7)		Attempt	1	0
17:36:07.033	Intra LTE-HO	(-/-) to (1600/356)	X2	Attempt	1	0
17:36:07.033	Intra LTE-HO	(-/-) to (1600/356)	X2	Success	0	1
17:36:08.433	Intra LTE-HO	(-/-) to (1600/30)	X2	Attempt	1	0
17:36:08.534	Intra LTE-HO	(-/-) to (1600/30)	X2	Success	0	1
17:36:18.746	Idle Mode Load Balancing	EUTRA(1600/7) EUTRA(275/6)		Attempt	1	0
17:37:31.603	Intra LTE-HO	(-/-) to (1600/31)	X2	Attempt	1	0
17:37:31.603	Intra LTE-HO	(-/-) to (1600/31)	X2	Success	0	1
17:37:37.326	Intra LTE-HO	(-/-) to (1600/15)	X2	Attempt	1	0
17:37:37.426	Intra LTE-HO	(-/-) to (1600/15)	X2	Success	0	1
17:38:06.643	Intra LTE-HO	(-/-) to (1600/84)	X2	Attempt	1	0
17:38:06.643	Intra LTE-HO	(-/-) to (1600/84)	X2	Success	0	1
17:40:19.242	Idle Mode Load Balancing	EUTRA(40490/7) EUTRA(40688/7)		Attempt	1	0

Figure 4: Event Statistics for Handover Information Log

For extracting information on serving cell parameters, we utilized the “Call Statistics” option, selected the “LTE Summary Table”, and further filtered the data by choosing the “Serving Cell” view. This yielded essential serving cell identifiers, including the Physical Cell ID (PCI) and Carrier-to-Interference-plus-Noise Ratio (CINR).

Serving Cell CINR		
M1 - 4	12.00/8.00 dB	--

Figure 5: LTE Summary Table

To obtain detailed radio measurement data, we accessed the “Message” tab, selected “Signaling Message”, and filtered for “Measurement Report” messages. These messages included Serving Cell RSRP, RSRQ, and corresponding Neighbor Cell RSRP, RSRQ, along with Neighbor Cell IDs, all timestamped for precise temporal alignment. Measurement reports play a crucial role in LTE networks, as they are fundamental to the handover decision-making process, enabling the eNodeB to evaluate surrounding cell conditions and initiate mobility procedures based on preconfigured event thresholds.

Time	UE-NET	Channel	ID	Message
17:09:16.237	←	DL DCCH	LTE	LTE ueCapabilityEnquiry
17:09:16.237	→	UL DCCH	LTE	LTE ueCapabilityInformation
17:09:16.237	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:16.334	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:16.334	→	UL DCCH	LTE	LTE measurementReport
17:09:16.934	→	UL DCCH	LTE	LTE measurementReport
17:09:16.934	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:16.935	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:17.334	→	UL DCCH	LTE	LTE measurementReport
17:09:17.639	→	UL DCCH	LTE	LTE measurementReport
17:09:17.643	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:17.739	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:17.739	←	BCCH:BCH	LTE	LTE MasterInformationBlock
17:09:17.739	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:17.740	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:17.740	←	BCCH:DL SCH	LTE	LTE systemInformationBlockType1 - MCC : 470(Bangladesh) / MNC : 1(Grameenpl
17:09:17.740	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:17.740	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:17.839	→	UL DCCH	LTE	LTE measurementReport
17:09:17.839	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:17.840	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:18.440	→	UL DCCH	LTE	LTE measurementReport
17:09:18.540	→	UL DCCH	LTE	LTE measurementReport
17:09:19.340	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:19.340	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:19.540	←	DL PCCH	LTE	LTE paging
17:09:20.740	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration
17:09:20.740	→	UL DCCH	LTE	LTE rrcConnectionReconfigurationComplete
17:09:20.840	←	DL PCCH	LTE	LTE paging
17:09:21.240	→	UL DCCH	LTE	LTE measurementReport
17:09:21.840	←	DL DCCH	LTE	LTE rrcConnectionReconfiguration

PC Timestamp: [2024 Oct 15 17:09:18.440]

Msg sequence : 43169

SUB_Command : 0x02(DM LTE Data Out Message)

Command_Type : 0x52

Time : 1913793651

nDataLength : 8

value UL-DCCH-Message :=

message c1 : measurementReport :

criticalExtensions c1 : measurementReport-r8 :

measResults

measId 5,

measResultPCell

rsrpResult 37,

rsrqResult 21

L3 Data in Hex :

08 02 25 54

Figure 6: LTE Summary Table

In LTE, Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) are critical physical layer metrics reported by the User Equipment (UE) in Layer 3 Measurement Reports, which are governed by the MeasConfig Information Element (IE). The S-Measure field within MeasConfig defines the RSRP threshold below which the UE initiates neighbor cell measurements. If S-Measure = 0, the UE is configured to perform neighbor measurements continuously, regardless of the serving cell’s RSRP level.

RSRP and RSRQ thresholds are quantized using RSRP-Range and RSRQ-Range IEs, respectively. The RSRP-Range IE supports integer values from 0 to 97, each corresponding to a 1 dB interval from below -140 dBm up to and beyond -44 dBm. The RSRQ-Range IE spans 0 to 34, mapping to standard RSRQ intervals. These quantized values appear in the Measurement Report under the signaling messages tab in XCAL-M software.

3.6 Data Preprocessing:

The objective of applying machine learning (ML) to the drive test dataset is to model and predict handover decisions based on real-world LTE mobility patterns. In this context, the handover serves as the target variable, while the number of handovers is used as the primary performance metric, with a focus on minimizing unnecessary handovers and ping-pong events through reinforcement learning-based optimization to enhance network stability and efficiency. To prepare the dataset for ML models, a series of preprocessing steps were

applied: missing RSRP and RSRQ values were interpolated manually based on serving cell IDs and timestamp continuity, data segments with hardware anomalies were excluded, and all non-LTE measurements were filtered out. Furthermore, the concept of Time-to-Trigger (TTT)—a standardized delay (set as 320 ms in our study) before a handover is executed—was incorporated. While raw handover labels were derived directly from Cell ID transitions in the Event Statistics log, the refined dataset assigns a handover label to those datapoints that contribute to a handover decision within the TTT window, thereby capturing the pre-handover context necessary for accurate model training.

Chapter 4

Overview of Handover Dataset

4.1 Dataset Structure:

The dataset is organized into five primary directories: Parent Datasets, Processed Datasets, DRM Files, Measurement Reports, and Event Statistics. Within the *Parent Datasets* folder, three subfolders—Raw Dataset_1, Raw Dataset_2, and Raw Dataset_3—contain 7, 2, and 3 CSV files respectively, representing the unprocessed drive test data. The *Processed Datasets* directory mirrors this structure with Processed Dataset_1, Processed Dataset_2, and Processed Dataset_3, each containing one CSV file with cleaned and ML-ready data. The preprocessing steps applied to these files are detailed in the following section.

The *Event Statistics* directory comprises three subfolders—Event Statistics_Dataset_1, Event Statistics_Dataset_2, and Event Statistics_Dataset_3—containing 7, 2, and 3 CSV files respectively. These files provide detailed LTE handover attempt and success records with timestamped events. Similarly, the *DRM Files* folder includes DRM Files_Dataset_1, DRM Files_Dataset_2, and DRM Files_Dataset_3, which contain the original XCAL-M log files (.drm format) with 7, 2, and 3 files respectively, enabling full reconstruction of the measurement sessions. The *Measurement Reports* directory features Measurement Reports_Dataset_1, Measurement Reports_Dataset_2, and Measurement Reports_Dataset_3, containing 7, 2, and 3 plain text files respectively, extracted from the Layer 3 signaling messages. Including Measurement Reports, Event Statistics, and DRM log files helps researchers trace the raw signaling and verify handover behavior, ensuring reproducibility and deeper analysis of real-world LTE mobility patterns.

The dataset contains 10 features and 1 target variable. All of them are numerically valued columns. There are 647 data points and 94 handover instances, indicating imbalance in the dataset. Table-1 shows the feature titles and the target variable.

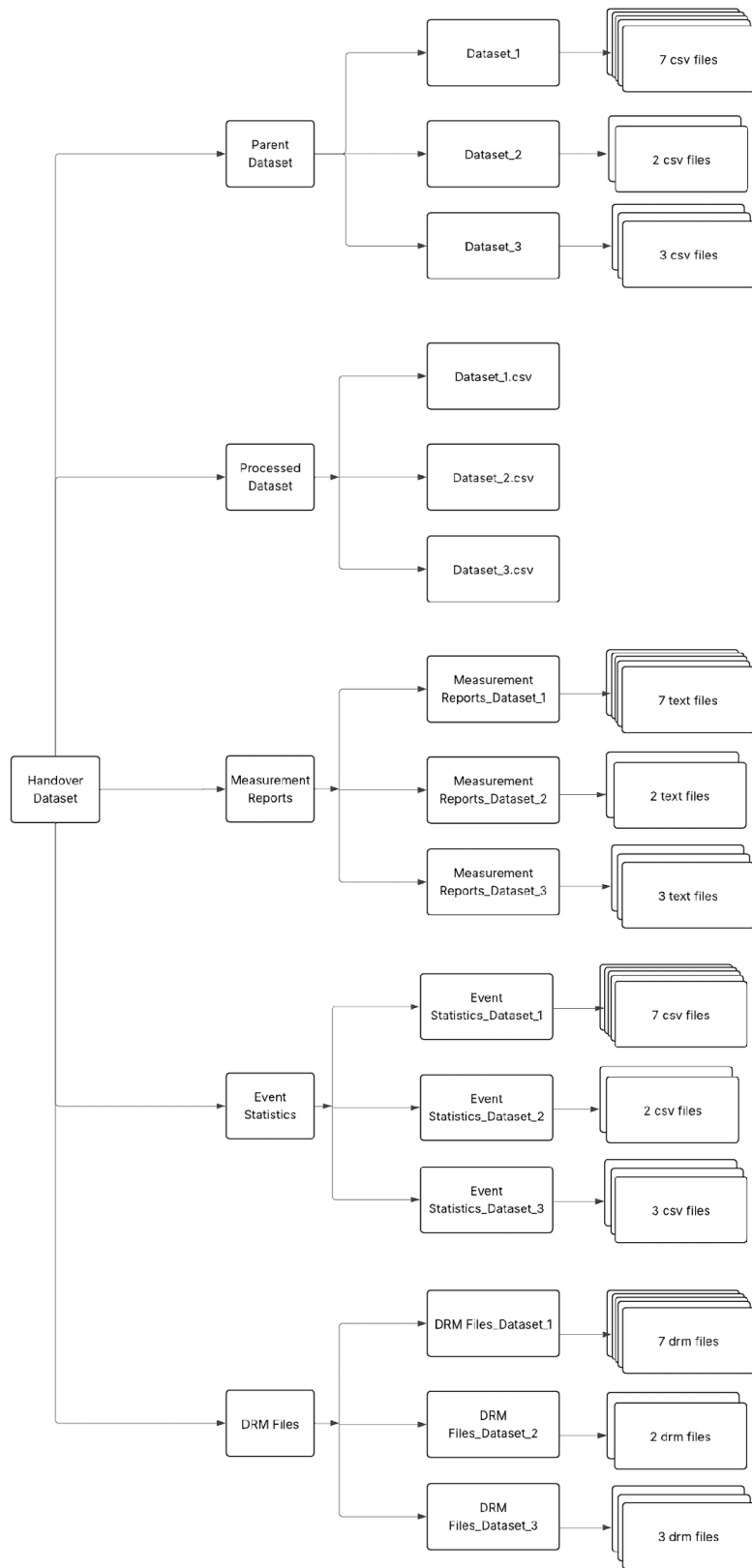


Figure 7: Dataset Structure

Table-1 shows the feature titles and the target variable.

Table 1 Dataset Columns

Features	Serving Cell Id
	Serving Cell RSRP
	Serving Cell RSRQ
	Serving Cell CINR
	Neighbor Cell Id
	Neighbor Cell RSRP
	Neighbor Cell RSRQ
	User Speed
	Latitude
	Longitude
Target Variable	Handover

Table 2 presents the parent dataset collected from the LTE drive test, while Table 3 shows the processed version of this dataset after cleaning and feature selection. Subsequently, Table 4 illustrates the handover instances observed with respect to the traveled distance during the test route.

Table 2 Parent Dataset:

Dataset Folder	Dataset Names	Number of Columns	Data Points
Dataset_1	File 1	11	132
	File 2	11	320
	File 3	11	74
	File 4	11	46
	File 5	11	243
	File 6	11	11
	File 7	11	654
Dataset_2	File 1	14	723
	File 2	14	798

Dataset_3	File 1	14	160
	File 2	14	339
	File 3	14	573

Table 3 Processed Dataset:

Dataset Names	Number of Columns	Data Points
Dataset 1	11	922
Dataset 2	14	592
Dataset 3	14	641

Table 4 Handover Instances with respect to travelled distance:

Time Period	Handover Count	Total Distance Travelled (km)	Distance Travelled per Handover in meter
Day 1	129	26.21	203
Day 2	94	25.54	268
Day 3	107	27.28	254

Table 2 summarizes the raw (parent) dataset collected from the LTE drive test, segmented into three folders based on distinct testing sessions or configurations. Each dataset comprises multiple files, with Dataset_1 containing seven files characterized by a consistent column count of 11, but a wide range of data points (from 11 to 654), indicating variability in recording durations or sampling intervals. Datasets 2 and 3 exhibit a slightly higher dimensionality with 14 columns, suggesting the inclusion of additional parameters latitude, longitude and user speed. The total number of data points across files provides a comprehensive foundation for further analysis but also reflects the heterogeneous nature of the raw data.

Table 3 presents the processed dataset, obtained after data cleaning, consolidation, and possibly filtering out noise or irrelevant records. The cleaned dataset shows a more balanced distribution of data points: Dataset_1 retains 922 points, while Datasets 2 and 3 hold 592 and 641 points, respectively. This reduction from the parent dataset reflects preprocessing efforts such as removing null entries, correcting inconsistencies, and ensuring uniform formatting—all critical for preparing data suitable for machine learning models.

Table 4 provides insights into handover behavior across three consecutive drive test days. Day 1 recorded 129 handovers over 26.21 km, resulting in an average of 203 meters per handover. Day 2 showed fewer handovers (95) over a similar distance, yielding 268 meters per handover, suggesting improved mobility handling or different route characteristics. Day 3 fell between the two with 107 handovers over 27.28 km, averaging 254 meters per handover. These variations reflect differences in network topology, traffic conditions, and signal propagation environments on different days. The metric "distance travelled per handover" serves as a critical performance indicator, where higher values typically imply more stable handover decisions and fewer disruptions, thus aiding in quantitative evaluation of mobility management efficiency.

4.2 Dataset Statistics:

Table 5, Table 6, and Table 7 present the descriptive statistics for Dataset_1, Dataset_2, and Dataset_3, respectively. These statistics summarize key wireless signal quality indicators such as Serving Cell RSRP, RSRQ, CINR, and their corresponding Neighbor Cell metrics. Dataset_2 and Dataset_3 also include speed as a contextual mobility parameter.

Table 5 Descriptive statistics of Dataset_1:

	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	922	922	922	922	922
mean	47.04446855	12.40997831	-2.023318872	45.64533623	9.252711497
std	13.03553886	7.225018978	10.80873338	11.61539961	7.744573122
min	13	0	-30	1	0
25%	42	7	-6.5	41	3
50%	48	13	-2.5	46	7
75%	54	17	2	53	14
max	78	51	31.5	77	45

Table 6 Descriptive statistics of Dataset_2:

	Speed (km/h)	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	592	592	592	592	592	592
mean	3.333044	52.12837838	11.19087838	-1.352195946	50.30152	7.787162
std	3.983686	9.307152358	6.049140385	5.856911567	9.122167	6.026897
min	0	21	0	-15	22	0
25%	0.445	48	7	-5	46	2.75
50%	1.22	53	11	-1.5	52	7
75%	5.4525	57	16	1.5	56	12
max	14.49	76	24	25	76	24

Table 7 Descriptive statistics of Dataset_3:

	Speed (km/h)	Serving Cell RSRP	Serving Cell RSRQ	Serving Cell CINR	NeighCell RSRP	NeighCell RSRQ
count	641	641	641	641	641	641
mean	23.4493	52.19032761	11.82371295	-1.06318	51.04680187	7.689547582
std	14.61242	10.66835127	7.061679108	8.310685	10.07895362	6.278338252

min	0	19	0	-30	18	0
25%	11	44	6	-4.5	44	3
50%	26	53	12	-1	51	7
75%	34	60	16	4	58	11

Table 5 presents the descriptive statistics for Dataset_1, which does not include user speed but provides insight into the variability of radio signal conditions. The Serving Cell RSRP has a mean of approximately 47 dBm, with values ranging from 13 dBm to 78 dBm, indicating diverse signal strengths experienced along the test route. The CINR (Carrier-to-Interference-plus-Noise Ratio) exhibits a wide range from -30 dB to 31.5 dB, reflecting fluctuations between highly degraded and very favorable radio environments. The RSRQ values also span a broad range (0 to 51), highlighting dynamic interference and load conditions.

Table 6 shows the statistics for Dataset_2, which includes speed information, with a mean speed of around 3.33 km/h, indicating primarily stationary or low-mobility conditions. Under these circumstances, the Serving Cell RSRP and CINR metrics are generally stronger and more stable, with average values of 52 dBm and -1.35 dB, respectively. The lower variability in signal quality metrics compared to Dataset_1 suggests improved link reliability and reduced handover frequency, aligning with the expected performance in static or pedestrian scenarios.

Table 7 corresponds to Dataset_3, where the mean speed is substantially higher at 23.45 km/h, representing vehicular mobility. The Serving Cell RSRP remains strong on average (~52.2 dBm), but the CINR and RSRQ metrics show greater dispersion, with CINR ranging from -30 dB to 25 dB and RSRQ up to 31. These broader signal variations under high-speed conditions suggest more frequent signal degradation and recovery phases, likely leading to increased handover events and signaling overhead. This dataset is therefore highly relevant for training mobility-aware handover prediction models under realistic, high-mobility scenarios.

4.3 Exploratory Data Analysis (EDA):

Figure 8, Figure 9, and Figure 10 present the correlation matrices for Dataset_1, Dataset_2, and Dataset_3, respectively. These matrices reveal the pairwise linear relationships between key wireless signal parameters.

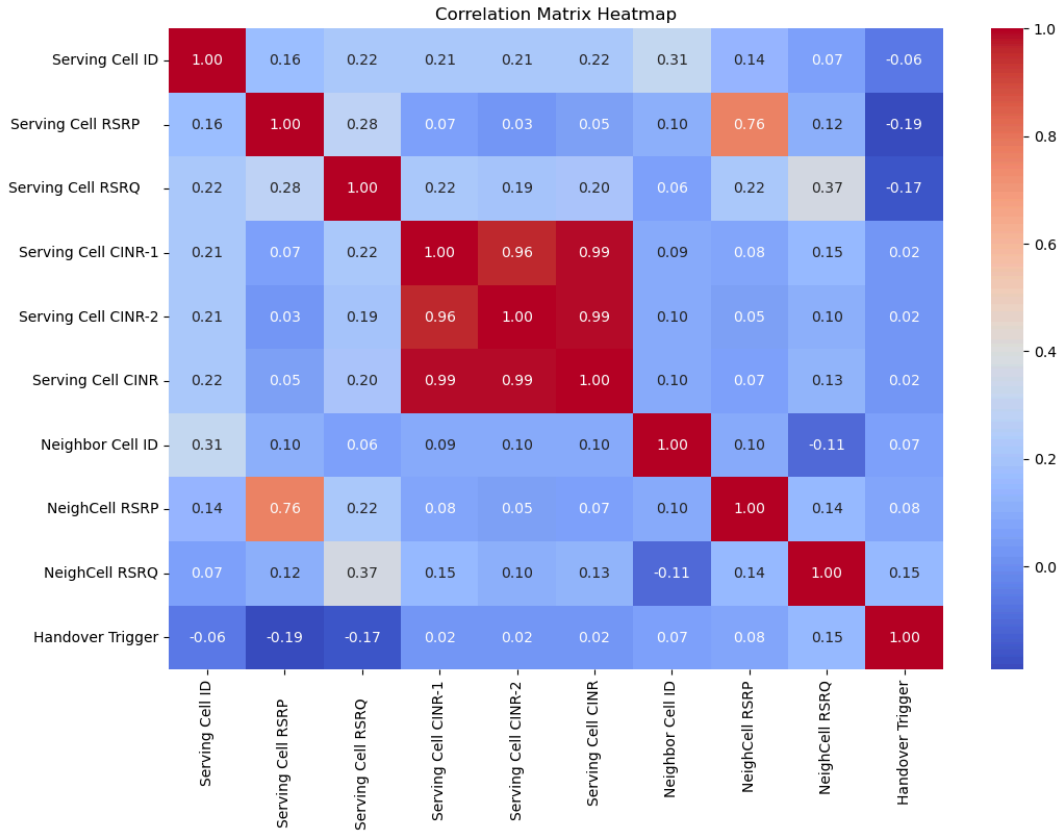


Figure 8: Correlation matrices for Dataset_1

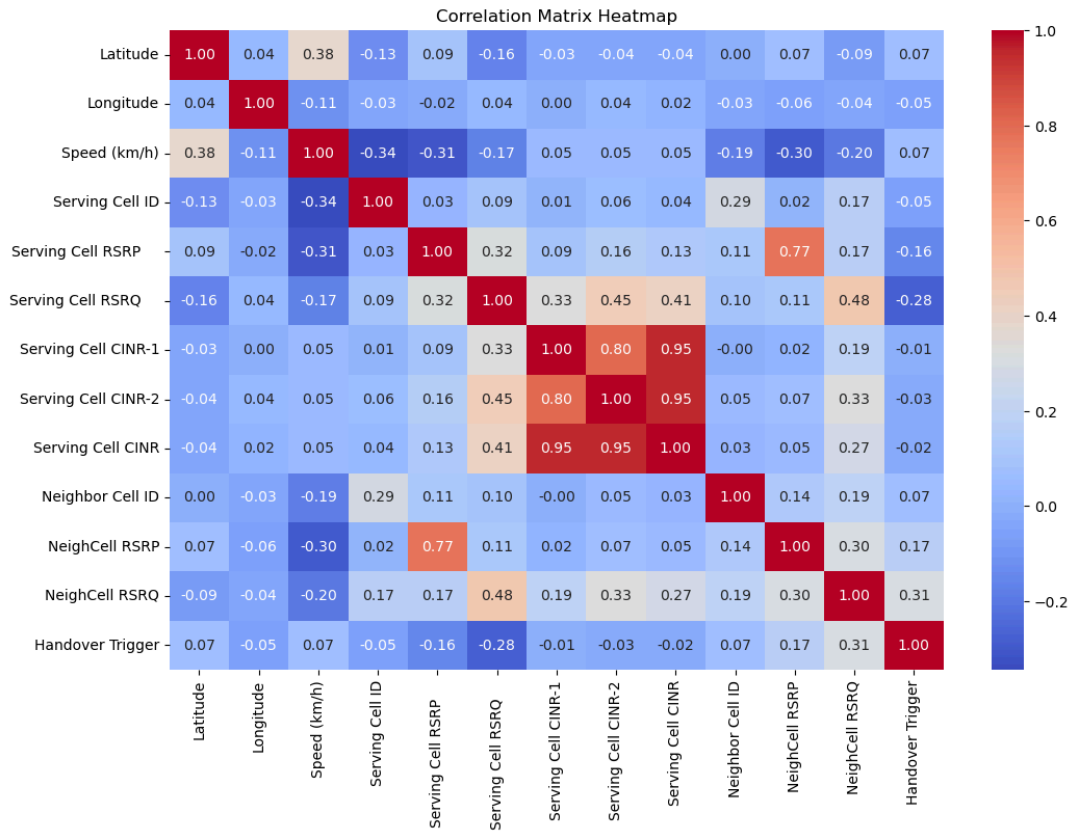


Figure 9: Correlation matrices for Dataset_2

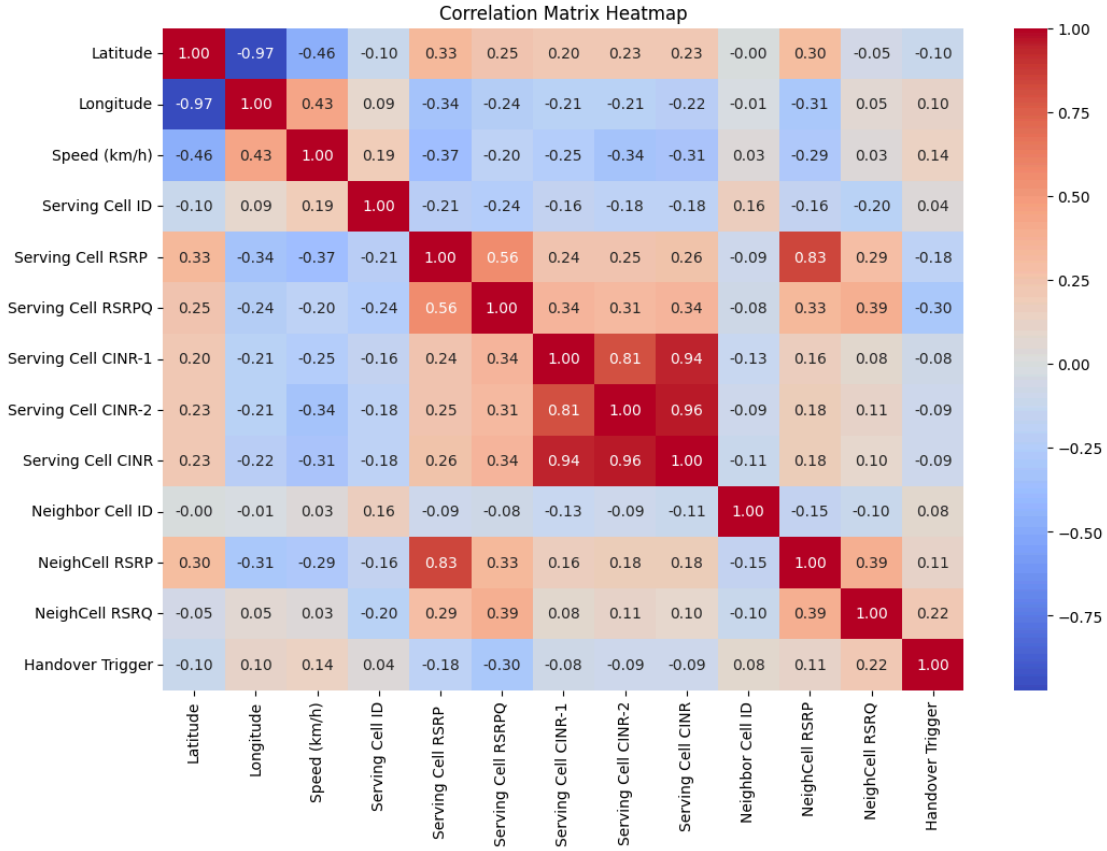


Figure 10: Correlation matrices for Dataset_3

Figure 10, Figure 11, and Figure 12 illustrate the handover frequency distribution across serving cells for Dataset_1, Dataset_2, and Dataset_3, respectively. These figures provide a visual representation of how frequently each serving cell initiated a handover during the drive test sessions.

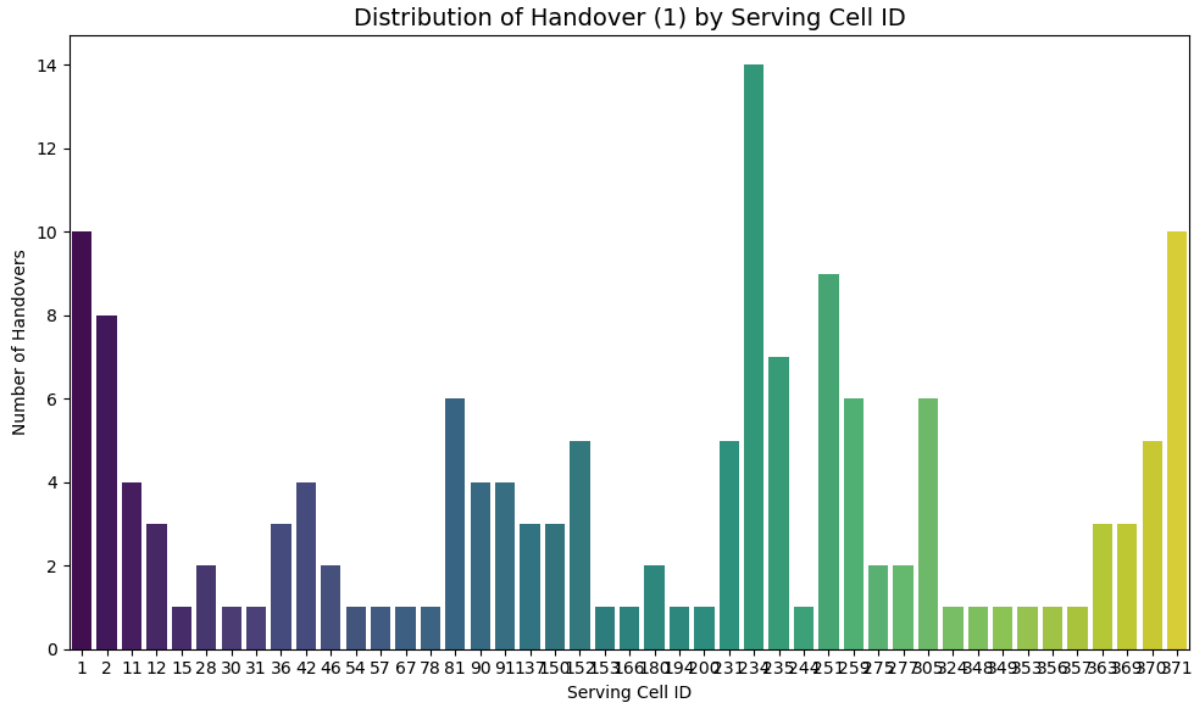


Figure 11: Handover Frequency over Serving Cells for Dataset_1

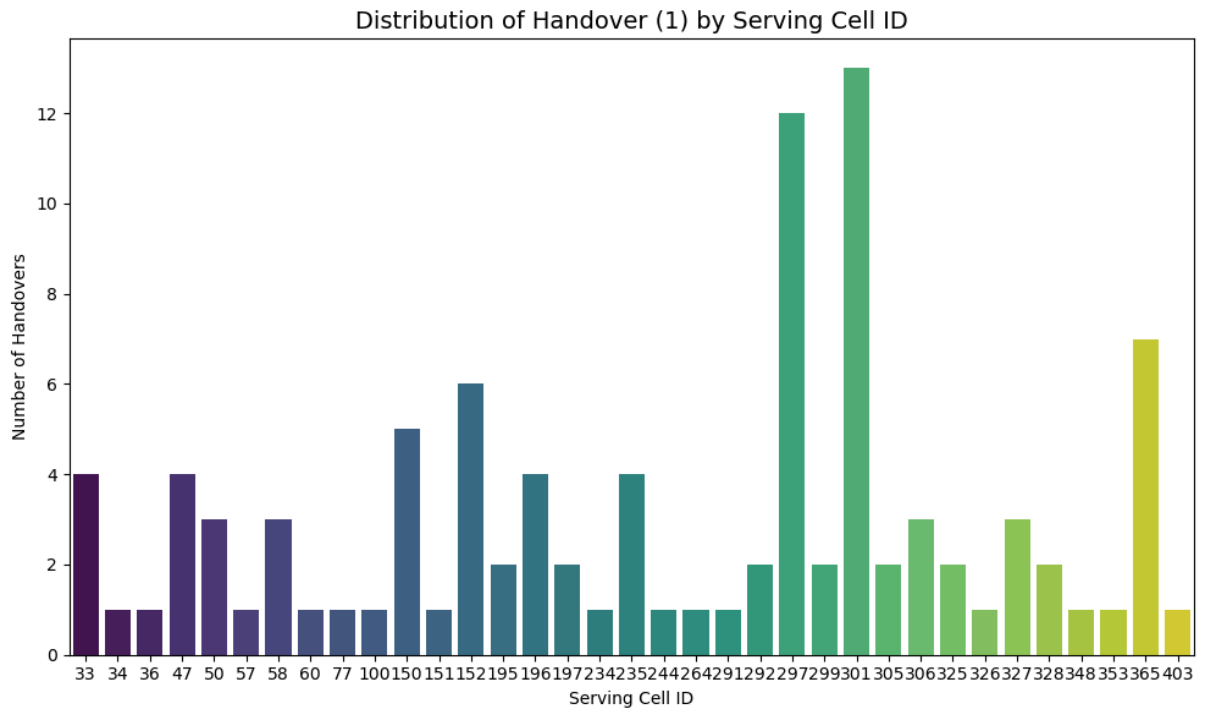


Figure 12: Handover Frequency over Serving Cells for Dataset_2

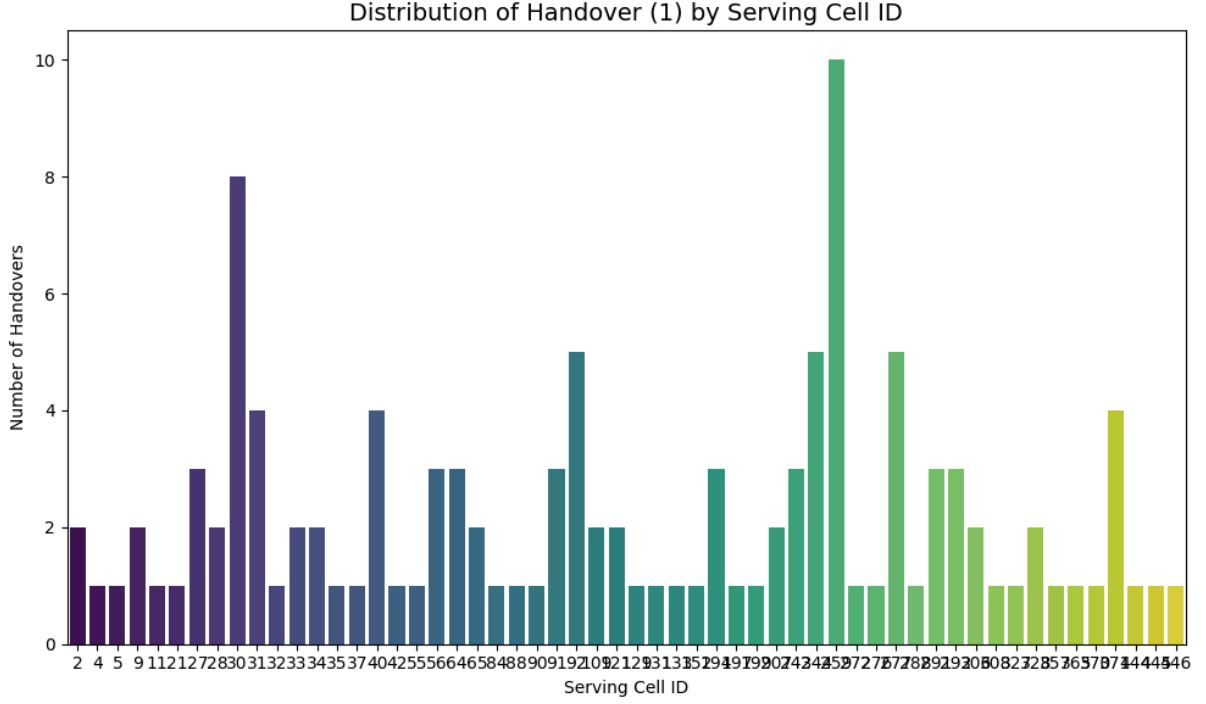
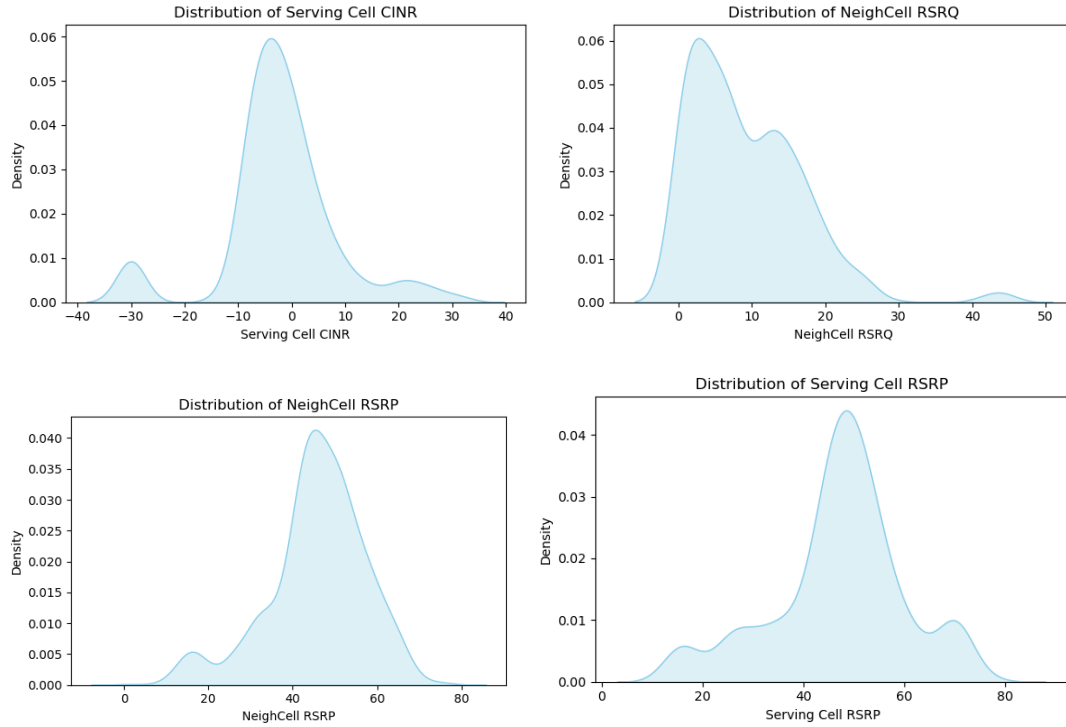


Figure 13: Handover Frequency over Serving Cells for Dataset_3

Figure 13, Figure 14, and Figure 15 illustrate the distribution of key network parameters—including UE speed, Serving Cell RSRP, RSRQ, CINR, and Neighbor Cell RSRP and RSRQ—for Dataset_1, Dataset_2, and Dataset_3, respectively. These figures provide insights into the statistical behavior of radio signal measurements and user mobility across varying conditions.



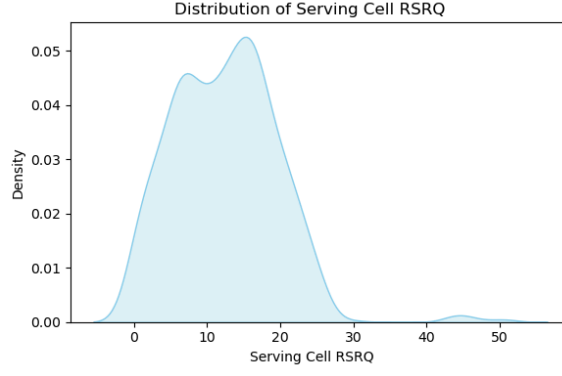


Figure 14: Distribution of Network Parameters of Dataset_1

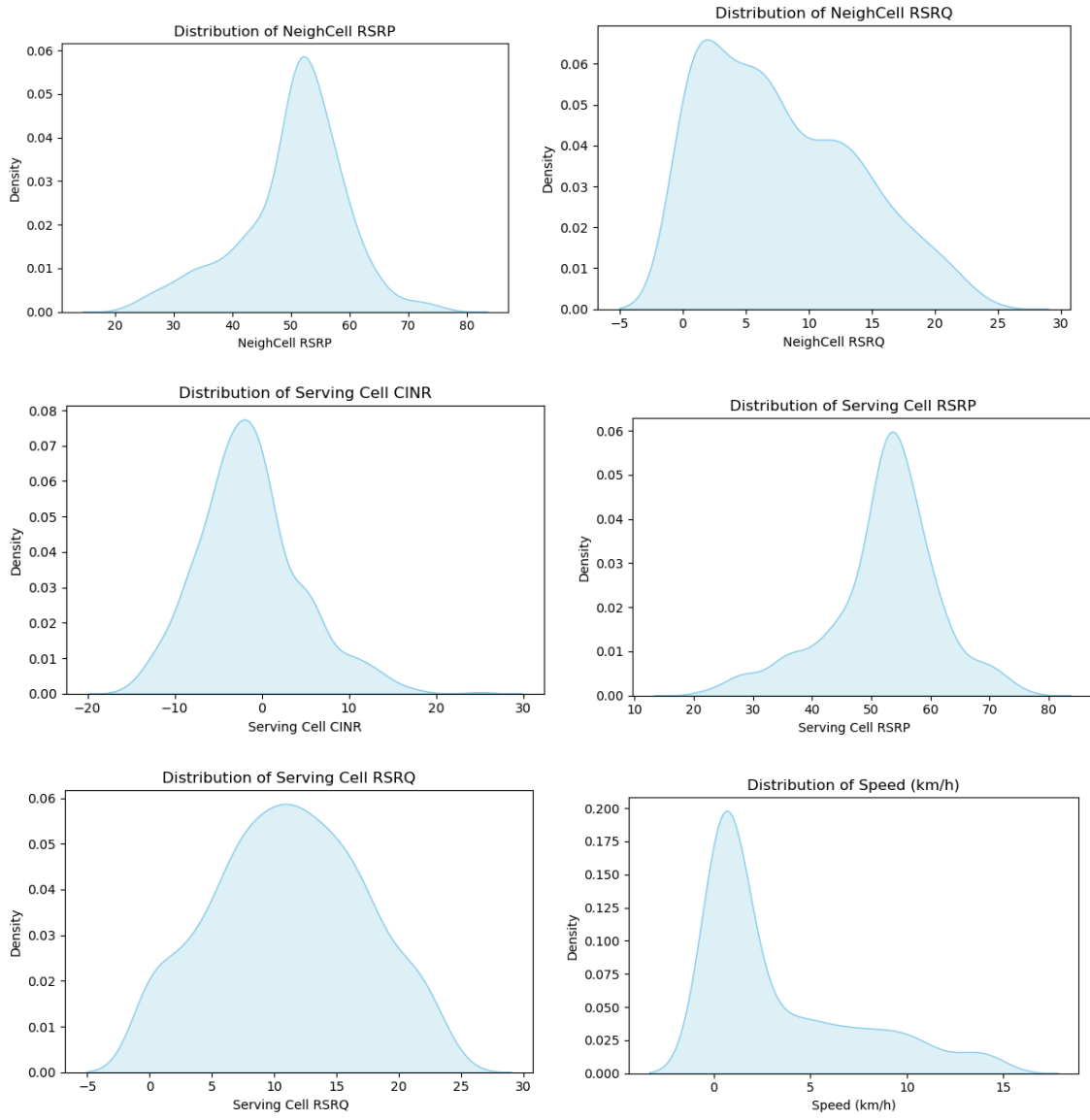


Figure 15: Distribution of Network Parameters of Dataset_2

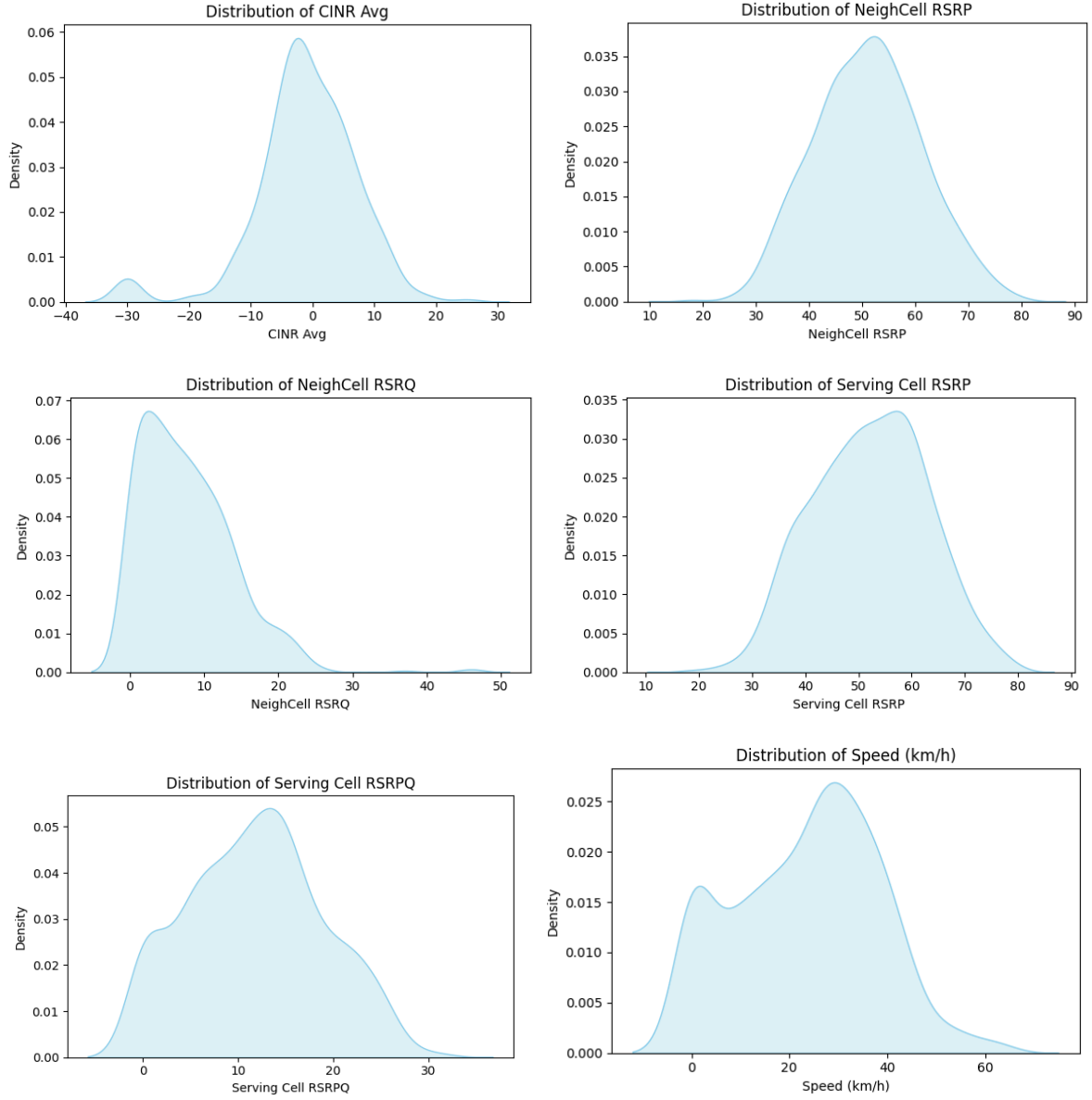


Figure 16: Distribution of Network Parameters of Dataset_3

The visual distributions indicate that the aforementioned parameters approximately follow a normal (Gaussian) distribution in each dataset. This is particularly evident in Serving Cell RSRP, RSRQ, and CINR, which display symmetric bell-shaped curves centered around their respective means. In Dataset_2 and Dataset_3, UE speed also exhibits a unimodal distribution with slight skewness due to the nature of user movement during the drive tests.

Figure 17 and Figure 18 shows the relation between RSRP and RSRQ values for both serving cell and neighbor cell in scatter plot.

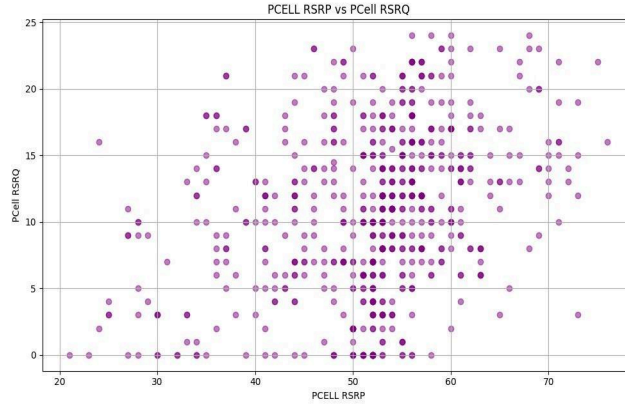


Figure 17: Scatter plot of Serving Cell RSRP and RSRQ

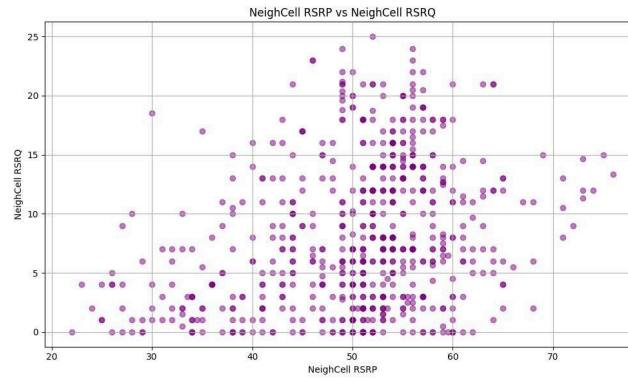


Figure 18: Scatter plot of Neighbor Cell RSRP and RSRQ

In the following chapters, we discuss the research aspects based on the dataset we generated. Our study focuses on three key aspects: Quality Signal Indicator (QSI), Handover Prediction, and Handover Optimization. We have selected Dataset_2 from the Processed Dataset folder for our experiment.

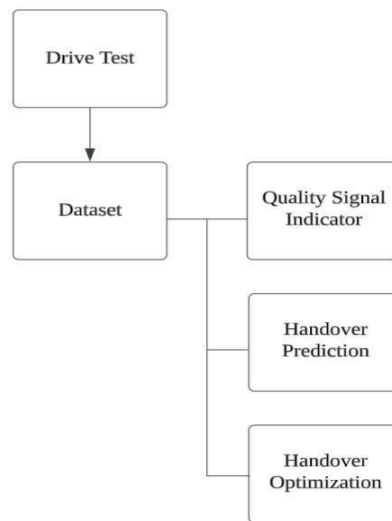


Figure 19: Thesis Overview

Chapter 5

Quality Signal Indicator

5.1 Signal Quality Labeling and Categorization

The dataset compiled for this study captures real-time radio signal quality indicators fundamental to cellular network performance assessment. These include reference signal received power (RSRP), reference signal received quality (RSRQ), and carrier-to-interference-plus-noise ratio (CINR), each measured for the serving cell. These parameters are widely recognized in 3GPP standards as primary indicators of downlink signal conditions and are instrumental in evaluating coverage, interference levels, and overall user experience within an LTE network.

To enable supervised learning-based model development, a manual threshold-based labeling scheme was employed to categorize the signal quality into four distinct classes: excellent, good, moderate, and poor. This categorization was guided by empirical domain knowledge and conventional telecom thresholds typically used in network optimization and planning processes. For instance, higher RSRP values generally correspond to stronger signal reception, while RSRQ and CINR provide insights into network load and interference environments.

This labeling approach constitutes a self-supervised learning methodology, wherein labels are derived from the input data itself using rule-based conditions rather than external annotation. Predefined thresholds were applied conditionally to each of the three signal metrics, and a composite rule was developed to assign one of the four quality labels to each data instance. For example, a sample exhibiting high RSRP, low RSRQ degradation, and a high CINR was labeled as "excellent", whereas a sample with significantly low RSRP and degraded RSRQ or CINR was labeled as "poor".

5.2 Model Training on Self Supervised Data

In this study, a comprehensive evaluation of nine machine learning classifiers was conducted to address a multiclass classification problem focused on wireless signal quality assessment. The classifiers employed include logistic regression, random forest, k-nearest neighbors (KNN), gradient boosting, XGBoost, AdaBoost, decision tree, naïve Bayes, and support vector machine (SVM). These algorithms were selected due to their proven effectiveness across a range of supervised learning tasks, as well as their complementary strengths in handling non-linearity, overfitting, interpretability, and scalability.

The input data was derived from a self-supervised signal quality labeling process using K-means clustering, ensuring that the labels represent a meaningful separation of network conditions based on data patterns rather than static thresholding. The resulting target variable, quality signal indicator (QSI), was categorized into four qualitative classes: excellent, good, moderate, and poor, based on principal wireless signal parameters.

The models were trained on key radio features that significantly influence user experience and network performance: reference signal received power (RSRP), reference signal received

quality (RSRQ), and signal-to-noise ratio (SNR). These features collectively characterize both the strength and quality of the downlink radio link and are widely used in radio network planning, optimization, and handover decision-making.

Each classifier was trained and validated using stratified sampling to preserve class distribution, ensuring fair and unbiased performance evaluation. Hyperparameter tuning was performed where appropriate to optimize learning efficiency and generalization. The goal was to build a robust, scalable, and interpretable predictive model that can accurately classify the signal quality class in various network scenarios—ranging from low-mobility urban environments to high-speed mobility contexts—thus supporting intelligent decision-making in handover optimization, quality of service (QoS) assurance, and coverage analysis within LTE or other advanced cellular systems.

5.3 Dimensionality Reduction Using Principal Component Analysis (PCA)

To improve computational efficiency and reduce feature redundancy, principal component analysis (PCA) was applied as a dimensionality reduction technique prior to clustering. PCA is a powerful linear transformation method that projects high-dimensional data onto a new coordinate system, where each axis—referred to as a principal component—is a linear combination of the original features and is orthogonal to the others. These components are ordered such that the first principal component captures the maximum possible variance in the dataset, the second captures the next highest variance under the constraint of orthogonality, and so on.

In the context of this study, the original feature set comprised multiple correlated wireless signal quality metrics, including serving cell RSRP, RSRQ, CINR, and neighbor cell RSRP/RSRQ, which often exhibit interdependencies due to radio propagation behavior and network configuration. PCA was employed to transform this multi-dimensional feature space into a set of uncorrelated principal components, thereby eliminating multicollinearity and emphasizing the most informative aspects of the data.

By retaining only the top k principal components (where k is chosen based on the cumulative explained variance criterion, typically $\geq 95\%$), the dimensionality of the dataset was significantly reduced without substantial loss of information. This not only accelerates the performance of subsequent clustering algorithms such as K-means by lowering the computational burden, but also improves the stability and separability of clusters by focusing on the most discriminative directions in the feature space.

Ultimately, PCA facilitated the creation of a more compact and representative dataset, which is crucial for large-scale network performance assessments and real-time handover optimization systems, where both speed and accuracy are paramount.

5.4 Clustering and Centroid Extraction

Following PCA, we visualized the signal categories using a scatter plot to analyze the distribution of signal quality clusters. From this plot, we extracted the centroids of each category, which served as the initial cluster centroids for K-means clustering.

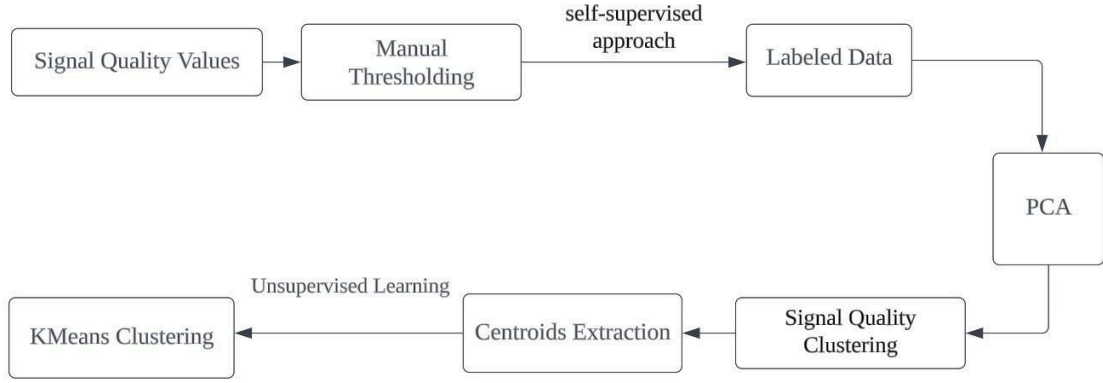


Figure 19: QSI workflow

5.6 Unsupervised Clustering with K-Means

To enhance the classification of signal quality, the K-means clustering algorithm, a widely used unsupervised machine learning technique, was implemented. Unlike traditional threshold-based methods that rely on predefined signal strength cutoffs (e.g., fixed RSRP or CINR ranges), K-means enables a more data-driven and adaptive classification by uncovering intrinsic groupings within the network measurement data.

The algorithm operates through a series of iterative steps:

1. **Initialization:** Instead of selecting centroids randomly, initial centroids were carefully extracted from visual observations of the scatter plots, ensuring a more informed starting point that reflects the natural distribution of the data.
2. **Cluster Assignment:** Each data point—representing a multi-dimensional feature vector comprising parameters such as Serving Cell RSRP, RSRQ, CINR, and Neighbor Cell metrics—is assigned to the cluster whose centroid is closest, typically measured using Euclidean distance in feature space. This step ensures that data points with similar signal characteristics are grouped together.
3. **Centroid Update:** The algorithm recalculates the centroids by computing the mean vector of all data points within each cluster. This dynamic adjustment enables the cluster centers to move toward regions of higher point density, progressively refining the clusters' representation of the underlying data.
4. **Convergence:** The above steps repeat iteratively until the centroids no longer exhibit significant positional changes between iterations. This indicates that the algorithm has converged to a stable configuration and that optimal clustering has been achieved.

The application of K-means clustering in this context proves effective in segregating signal quality levels based on actual data patterns rather than relying on static or arbitrary thresholds. This not only enhances the granularity and reliability of classification but also improves the model's ability to generalize across different environments.

Chapter 6

Handover Prediction Model

6.1 Imbalance Handling and Dataset Preprocessing

Due to significant class imbalance in the dataset, we designed a robust handover prediction architecture to enhance model performance. The dataset was initially split into training and testing sets, with further partitioning of the training set into majority (non-handover) and minority (handover) subsets. To address imbalance, we applied Random Undersampling (RUS) for the majority class and Synthetic Minority Over-sampling Technique (SMOTE) for the minority class.

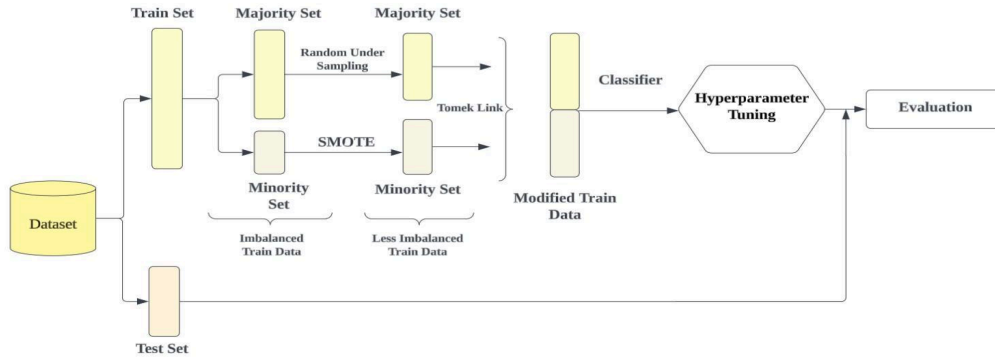


Figure 20: Proposed Architecture

Random Undersampling (RUS): This technique randomly removes samples from the majority class, reducing over-representation and preventing classifier bias.

SMOTE (Synthetic Minority Over-sampling Technique): SMOTE synthetically generates new minority class instances by interpolating between existing samples. It selects a minority sample, finds its nearest neighbors, and generates synthetic points along the feature space to improve class balance.

After rebalancing, we applied Tomek Links to further refine the dataset.

Tomek Link-Based Noise Reduction: Tomek Links are data pairs where one sample belongs to the majority class and the other to the minority class, with minimal inter-class distance. Removing the majority class instances in these pairs enhances class separation, improving classifier performance by reducing borderline noise.

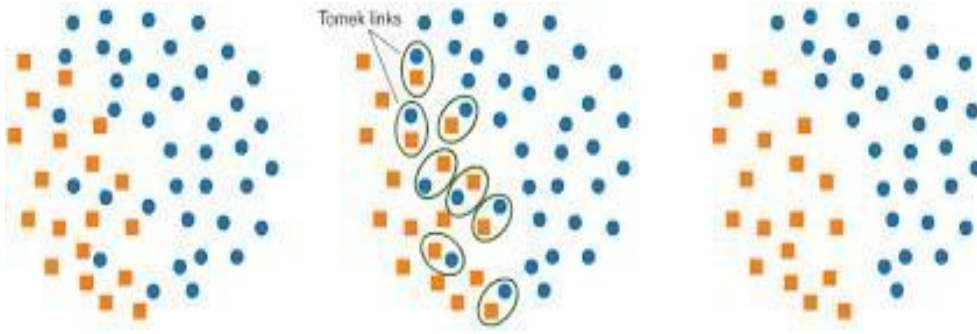


Figure 21: Tomek Link functionality (source: Kaggle)

6.2 Model Training

In this study, a total of ten supervised classification models were implemented and rigorously evaluated to predict handover decisions based on signal quality metrics. The classifiers included logistic regression, decision tree, random forest, k-nearest neighbors (k-NN), naïve Bayes, support vector machine (SVM), AdaBoost, gradient boosting, CatBoost, and XGBoost. These models were selected to represent a diverse set of algorithmic paradigms—ranging from linear models and probabilistic classifiers to ensemble learning methods and gradient-based boosting techniques—providing a comprehensive analysis of their suitability for the handover prediction task.

Each model was subjected to an extensive hyperparameter tuning process to ensure optimal configuration. Grid search and random search methods were employed to explore the parameter space systematically and stochastically, respectively. During this process, the models were cross-validated using stratified k-fold validation to maintain class balance and ensure robustness across folds. Performance was evaluated using multiple classification metrics, including accuracy, precision, recall, and F1-score, to capture different dimensions of model effectiveness—particularly in scenarios where class imbalance may affect metric interpretation.

Upon completion of the tuning and training phases, all models were evaluated on a reserved test set comprising unseen data. This final evaluation step was essential for estimating each model's generalization ability and determining its practical applicability to real-world network scenarios. The performance of each classifier was also benchmarked against a simple baseline model—such as a dummy classifier predicting the majority class—to contextualize the results and highlight the performance gains achieved through learning.

The comprehensive evaluation allowed for an informed comparison of model performance, ultimately identifying the most effective classifier for handover prediction based on real-time radio environment measurements. This process contributes to the broader goal of developing intelligent, data-driven mobility management solutions in next-generation wireless networks.

Chapter 7

Handover Optimization

7.1 Hybrid Reinforcement Learning Framework

To optimize handover decisions, we implemented a hybrid Q-learning model that integrates Logistic Regression-based action selection with traditional Q-learning. This approach balances model-driven decision-making with reinforcement learning to minimize unnecessary handovers while maintaining signal quality.

State, Action, and Reward Definition

In the Q-learning framework:

State (S_t): Each state represents a data point containing real-time RSRP, RSRQ, and CINR values.

Action (a_t): The agent (User Equipment) can choose between handover or non-handover.

Reward Function (R_t): The agent receives a positive reward for an optimal handover and a negative reward (penalty) for an unnecessary or failed handover. The reward function is designed to minimize frequent handovers while maintaining network stability.

Table 8: Reward Function:

Condition	Action	Reward/Penalty	Explanation
Action = 1 (Handover)	Handover	-	Decision to perform a handover
NeighCell RSRP $\leq$ PCell RSRP or NeighCell RSRQ $\leq$ PCell RSRQ	Handover	-3	Penalty for poor handover (worse neighbor)
NeighCell RSRP $>$ PCell RSRP and NeighCell RSRQ $>$ PCell RSRQ	Handover	+4	Reward for selecting a better neighboring cell
CINR $<$ 5 after handover	Handover	-1	Extra penalty for poor signal quality after handover
CINR $>$ 20 after handover	Handover	+3	Reward for achieving high signal quality after handover

Action = 0 (No Handover)	No Handover	-	Decision to stay connected to the current cell
CINR > 15 without handover	No Handover	+3	Reward for maintaining good connection quality
CINR ≤ 15 without handover	No Handover	+1	Mild reward for staying connected with moderate quality

Exploration-Exploitation Strategy

Reinforcement learning requires a balance between exploration (trying new actions) and exploitation (leveraging learned actions). Instead of using random action selection for exploration, we incorporated Logistic Regression-based action selection, making the model a hybrid Q-learning approach.

Early-stage learning: Higher exploration rate ensures the agent samples different scenarios.

Exploration decay: Over time, exploration decreases, allowing the agent to exploit learned policies from the Q-table.

Action selection via Logistic Regression: Instead of choosing actions randomly during exploration, we use a Logistic Regression model to predict the optimal action, providing a structured and informed decision-making process.

7.2 Q-Table Update and Policy Learning

The Q-learning update rule is applied iteratively:

$Q(S_t, a_t) \leftarrow Q(S_t, a_t) + \alpha [R_t + \beta \max_{\mathbf{a}} Q(S_{t+1}, \mathbf{a}) - Q(S_t, a_t)]$ where: α is the learning rate controlling the update step. β is the discount factor, determining the importance of future rewards. $\max Q(S_{t+1}, \mathbf{a})$ represents the best expected reward from the next state.

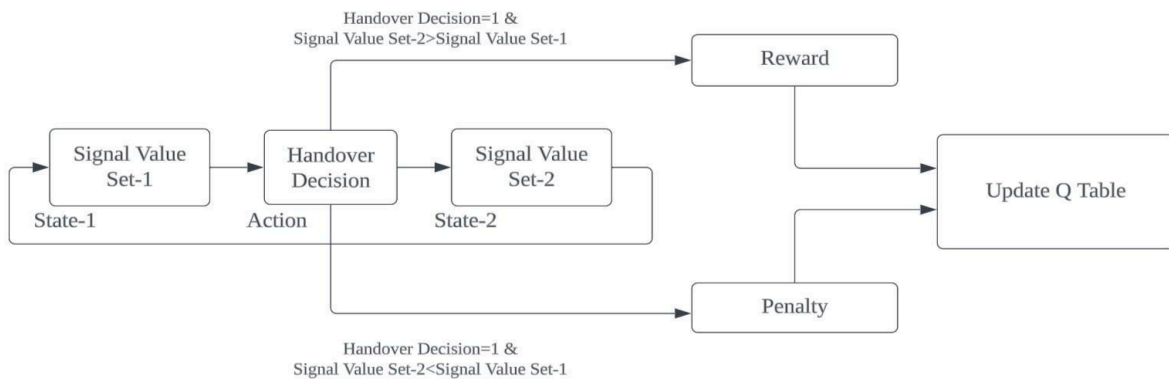


Figure 21: Q learning: Training Phase

By continuously updating the Q-table, the model refines its handover decisions, learning an optimal policy over time.

7.3 Evaluation and Handover Optimization

After training, the Q-table was evaluated to assess its impact on handover efficiency. The primary optimization parameter was the number of handovers, which was significantly reduced while ensuring seamless connectivity. This approach enables adaptive, data-driven handover optimization, reducing unnecessary transitions and enhancing network performance.

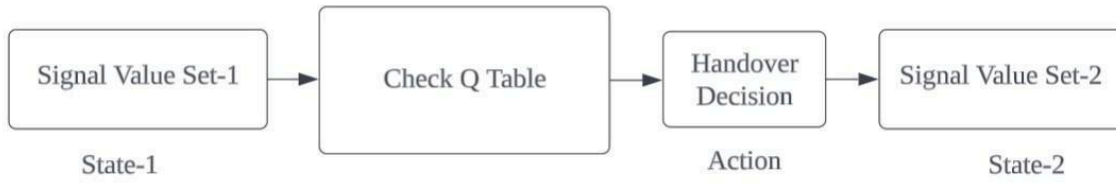


Figure 22: Q learning: Evaluation Phase

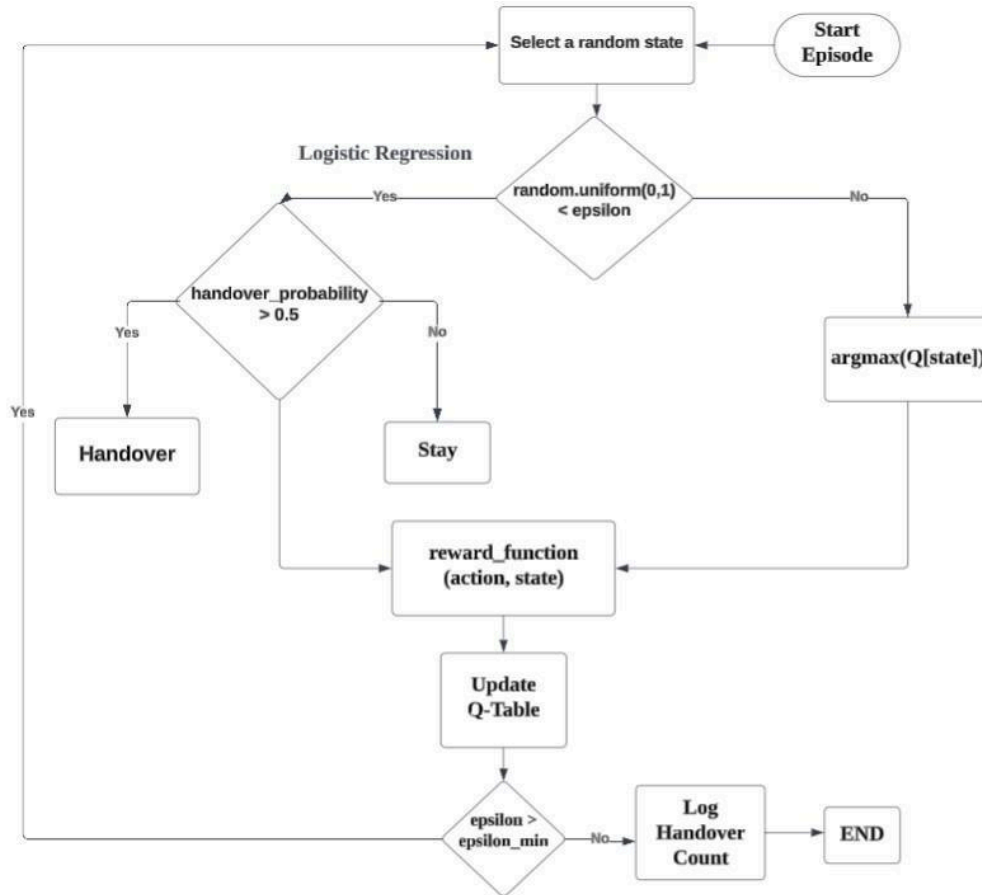


Figure 23: Flow Chart of the proposed algorithm

Chapter 8

Results

This section presents a comprehensive evaluation of the proposed model based on key performance metrics and experimental results. The analysis is structured into three main subsections: Quality Signal Indicator (QSI), handover classification, and handover optimization using hybrid Q-learning. First, the QSI assessment provides insights into signal quality distribution and its correlation with handover events. Next, the handover classification results are examined, comparing model performance against baseline approaches. Finally, the effectiveness of hybrid Q-learning in optimizing handover decisions is evaluated, demonstrating its impact on network performance.

9.1 Experimental result evaluation of QSI

For QSI analysis, we need only 3 features: serving cell RSRP, serving cell RSRQ and CINR. As per the methodology previously described in 4.1 section, first we performed self supervised learning and labeled the unlabeled data. We categorize them into 4 groups: Excellent, Good, Moderate and Poor. Table-9 shows the threshold mapping for the signal quality data.

Table 9: Threshold Mapping

Category	Threshold Quantile Percentage
Excellent	>90%
Good	>50%
Moderate	>15%
Poor	<15%

Figure 24 presents a comparison of the accuracy of the nine models evaluated in this study. Among the classifiers, Logistic Regression and Support Vector Machine (SVM) achieved the highest accuracy, demonstrating superior performance in predicting Quality Signal Indicator (QSI) categories. In contrast, AdaBoost showed the poorest performance, with notably lower accuracy compared to the other models. This comparison highlights the varying effectiveness of the classifiers in handling multiclass classification tasks for signal quality prediction.

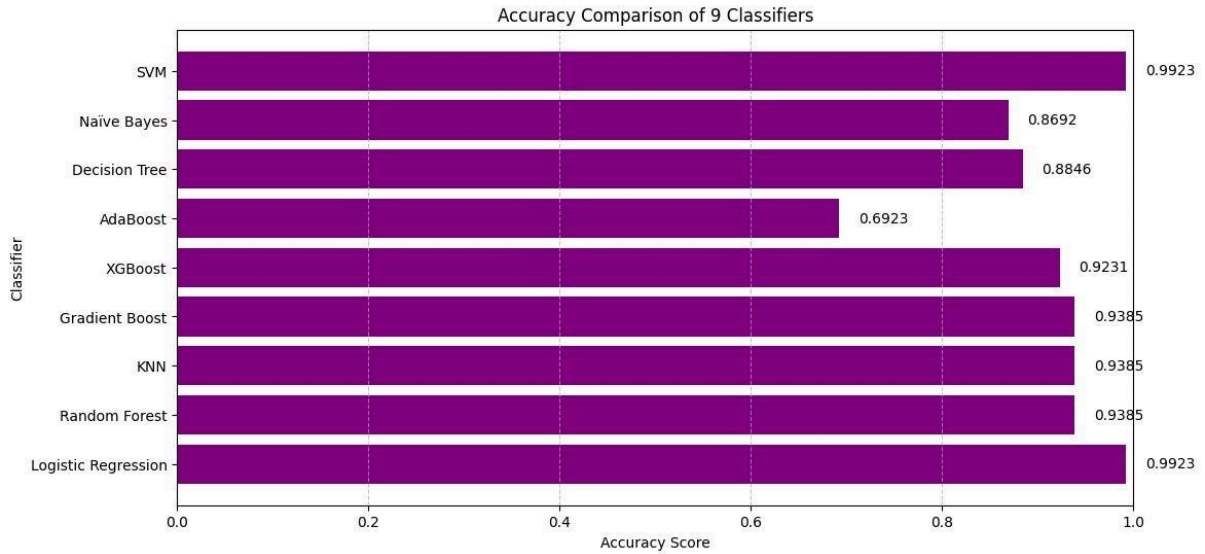


Figure 24: QSI Model Accuracy

We applied Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while preserving its key variance. Following the PCA transformation, we performed clustering to group similar data points based on their intrinsic patterns. The results were visualized using a scatter plot to illustrate the distribution and separability of the identified clusters. Figure 25 presents the scatter plot, highlighting the clustered categories and their relationships in the transformed feature space.

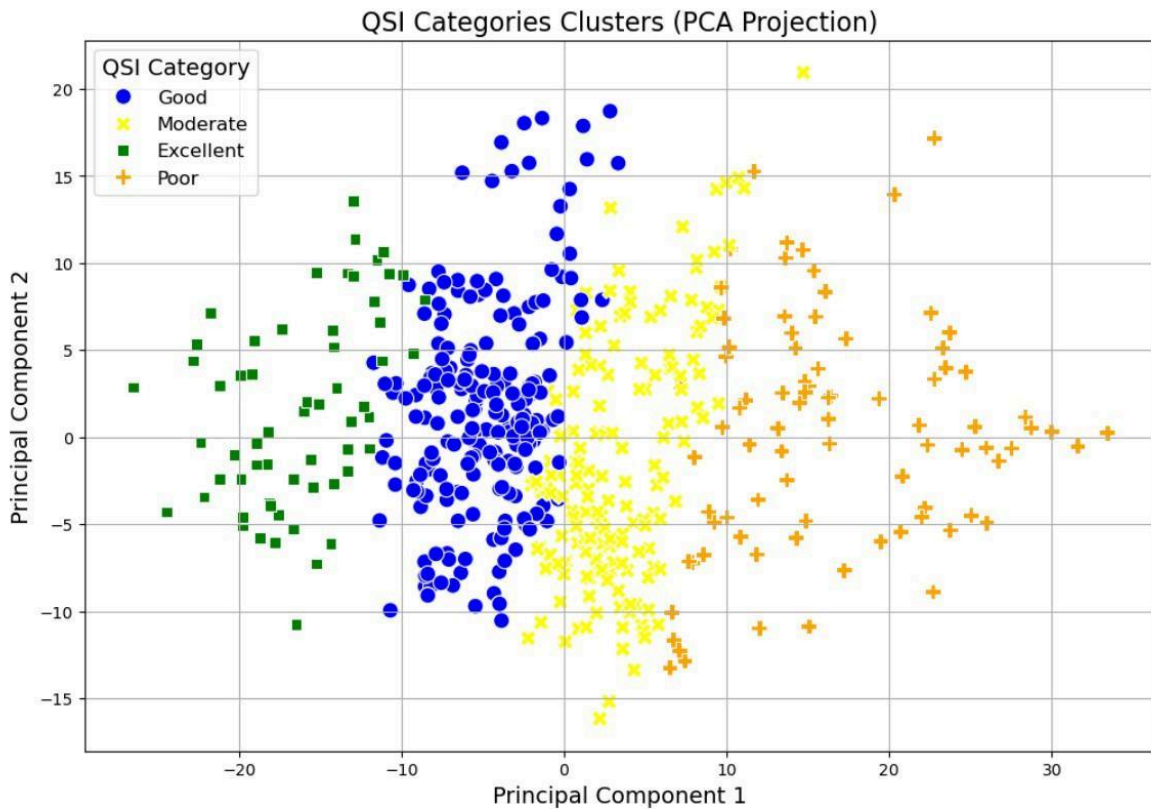


Figure 25 : QSI Clustering using Predefined Labeling

Now we extracted the centroids from this scatter plot of each category. Table-4 shows the PCA axis positions of the centroids.

Table 10: Centroid Positions

QSI_Category	PCA-1	PCA-2
Excellent	-16.042605	1.767194
Good	-4.831206	1.473416
Moderate	3.20405	-2.391944
Poor	16.350566	0.374912

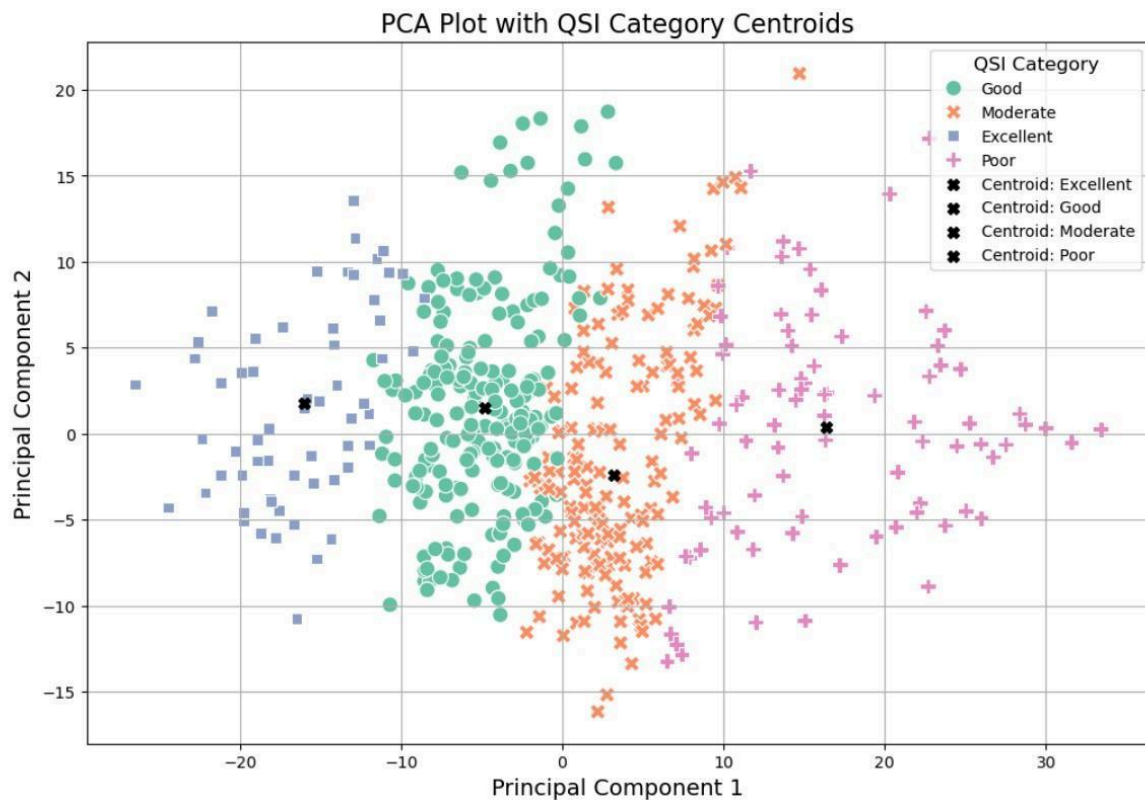


Figure 26: QSI clustering with centroids

Now, we employed K-Means clustering, an unsupervised machine learning algorithm, to classify Quality Signal Indicator (QSI) categories based on predefined centroids. The centroids were manually initialized using domain-specific knowledge, representing four distinct QSI categories: Excellent, Good, Moderate, and Poor. By specifying these centroids as the initial input, we ensured a guided clustering process that aligns with real-world network conditions. The K-Means algorithm was then executed with four clusters, a fixed initialization, and a maximum of 200 iterations, optimizing cluster assignments while maintaining consistency. Finally, the obtained cluster labels were mapped to their corresponding QSI categories, enabling a structured classification of signal quality levels in the dataset.

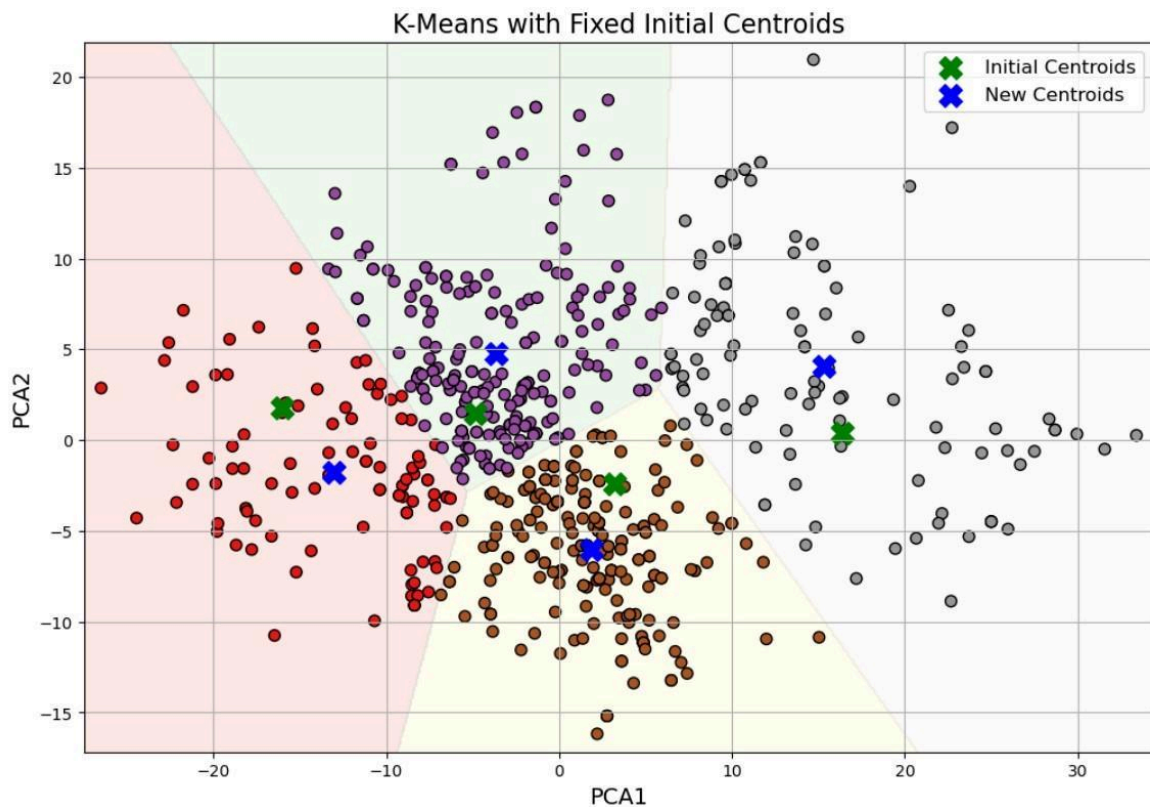


Figure 27: QSI Clustering using K-Means algorithm

9.2 Experimental result evaluation of Handover Prediction:

Evaluation Metrics Evaluation metrics are quantitative measures used to assess the performance of a machine learning model. These metrics help determine how well a model

makes predictions and whether it meets the desired objectives. The choice of evaluation metric depends on the type of problem being solved, such as classification, regression, or clustering.

Precision

Precision measures how many of the instances predicted as positive are actually positive. It is calculated as $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$, where TP (True Positives) are correctly predicted positive cases, and FP (False Positives) are incorrectly predicted positives. A high precision means fewer false positives, making it crucial in applications where false alarms are costly, such as medical diagnoses or fraud detection.

Recall

Recall, also known as sensitivity, measures how well the model captures actual positive cases. It is defined as $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$, where FN (False Negatives) are actual positives that were incorrectly classified as negative. A high recall ensures that most positive cases are identified, which is important in applications like disease screening, where missing a positive case can have severe consequences.

F1-Score

F1-score is the harmonic mean of precision and recall, balancing both metrics. It is calculated as $\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. This metric is particularly useful when there is an imbalance in the dataset, as it provides a single value that accounts for both false positives and false negatives. A high F1-score indicates that the model maintains a good trade-off between precision and recall.

Accuracy

Accuracy is a commonly used metric that measures the overall correctness of the model's predictions. It is defined as $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$, where TN (True Negatives) are correctly predicted negative cases. While accuracy is useful for balanced datasets, it can be misleading in imbalanced datasets where one class dominates. To evaluate the performance of the handover prediction model, we employed ten different classification algorithms, ensuring a comprehensive assessment of predictive accuracy. The proposed model architecture, as detailed in Section 4.2, was designed to enhance classification efficiency by incorporating imbalanced dataset handling techniques. The following figures illustrate a comparative analysis of performance metrics, including accuracy, precision, recall, and F1-score, between the baseline model and the proposed model. This evaluation provides 34 35 valuable insights into the effectiveness of the proposed approach in improving handover prediction outcomes.

Performance Comparison of Logistic Regression: Proposed Model vs. Baseline Model

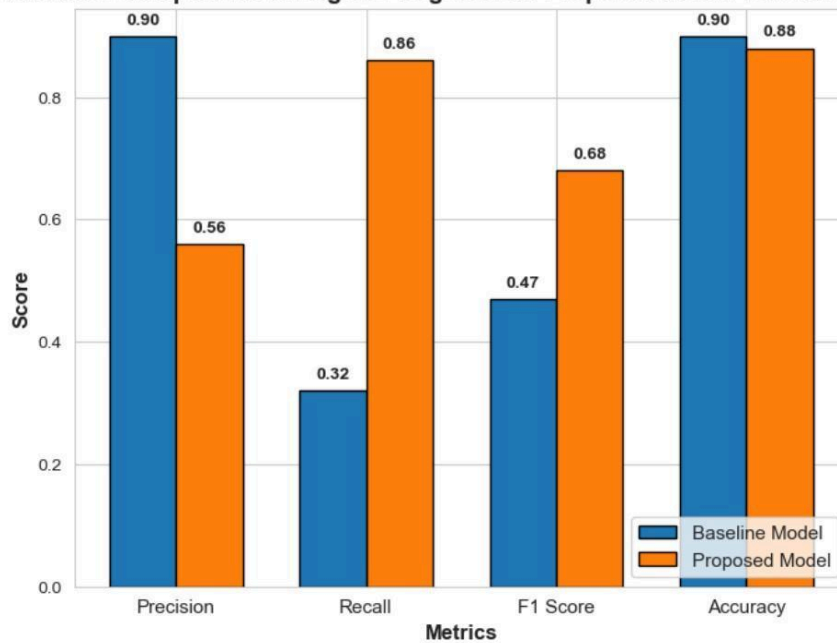


Figure 28: Logistic regression

Performance Comparison of Decision Tree: Proposed Model vs. Baseline Model

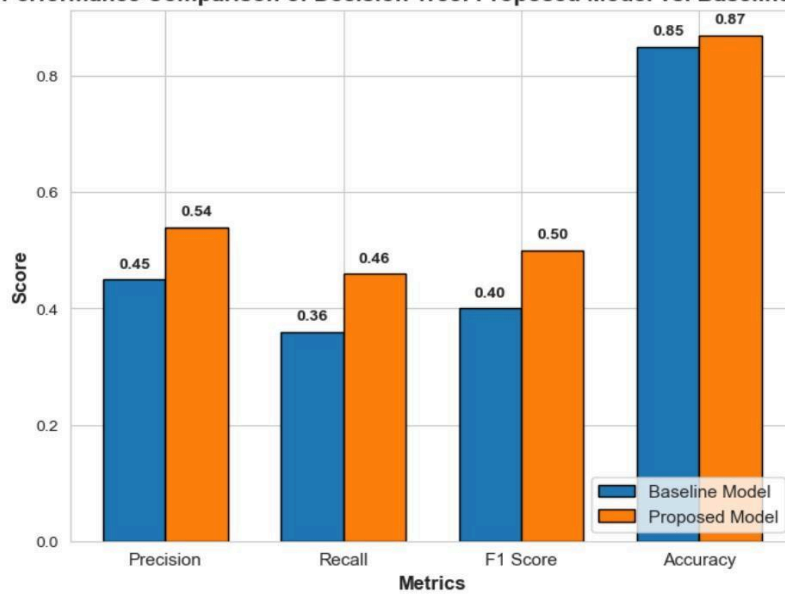


Figure 29: Decision tree

Performance Comparison of Random Forest: Proposed Model vs. Baseline Model

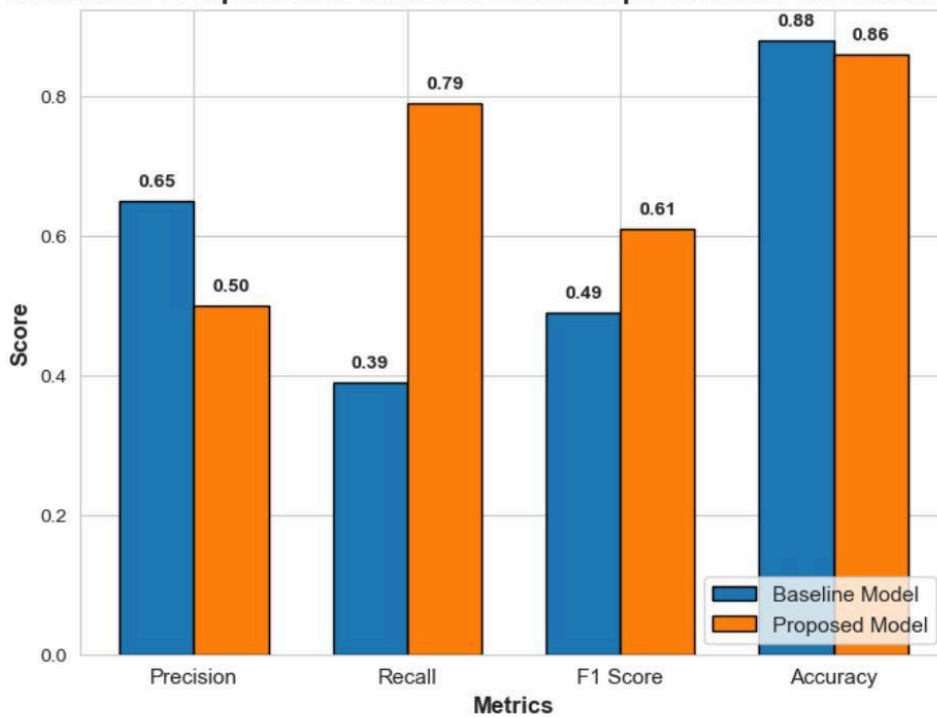


Figure 30: Random Forest

Performance Comparison of Naive Bayes: Proposed Model vs. Baseline Model

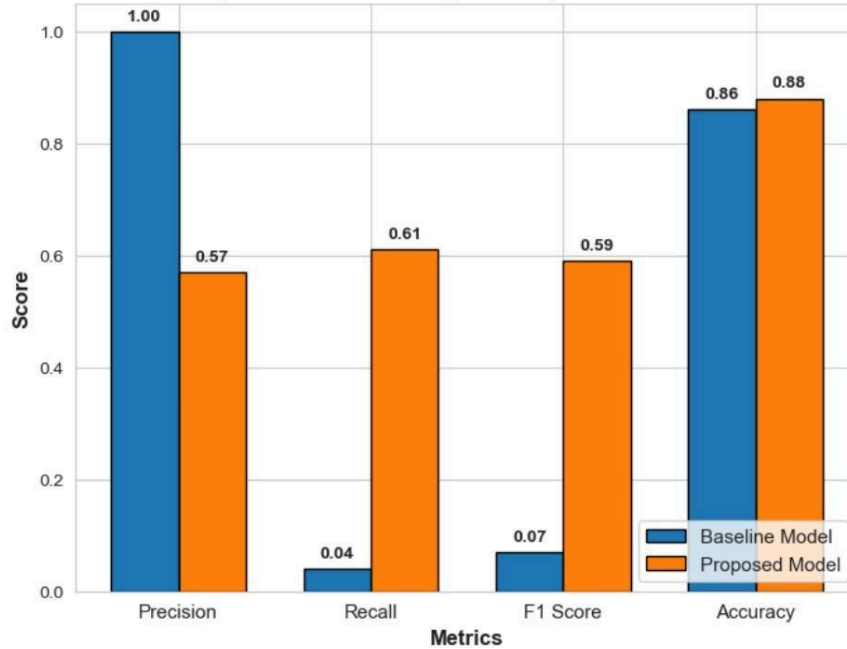


Figure 31: Naive Bayes.

Performance Comparison of K-nearest neighbor: Proposed Model vs. Baseline Model

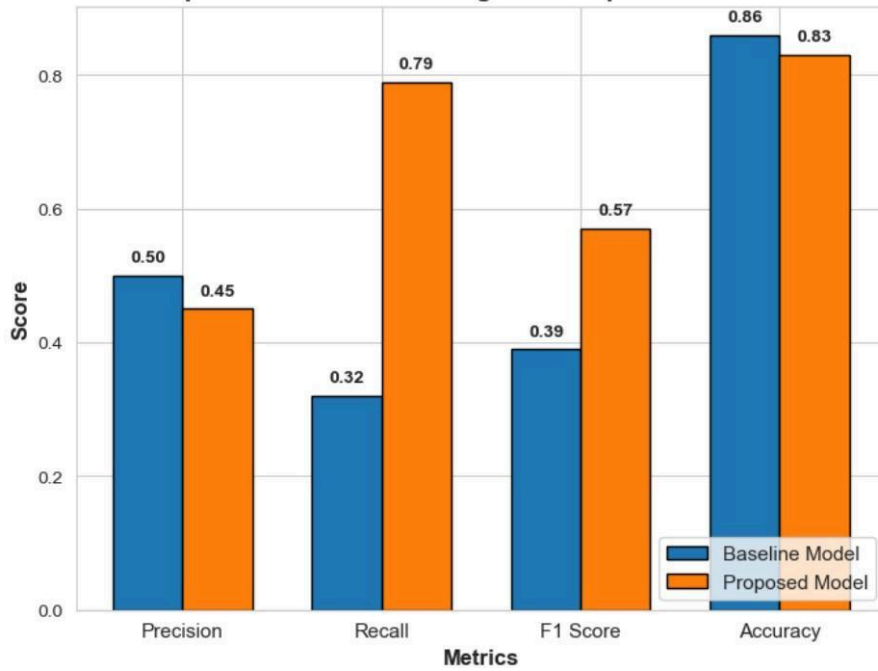


Figure 32: K-nearest neighbor

Performance Comparison of ADABOOST: Proposed Model vs. Baseline Model

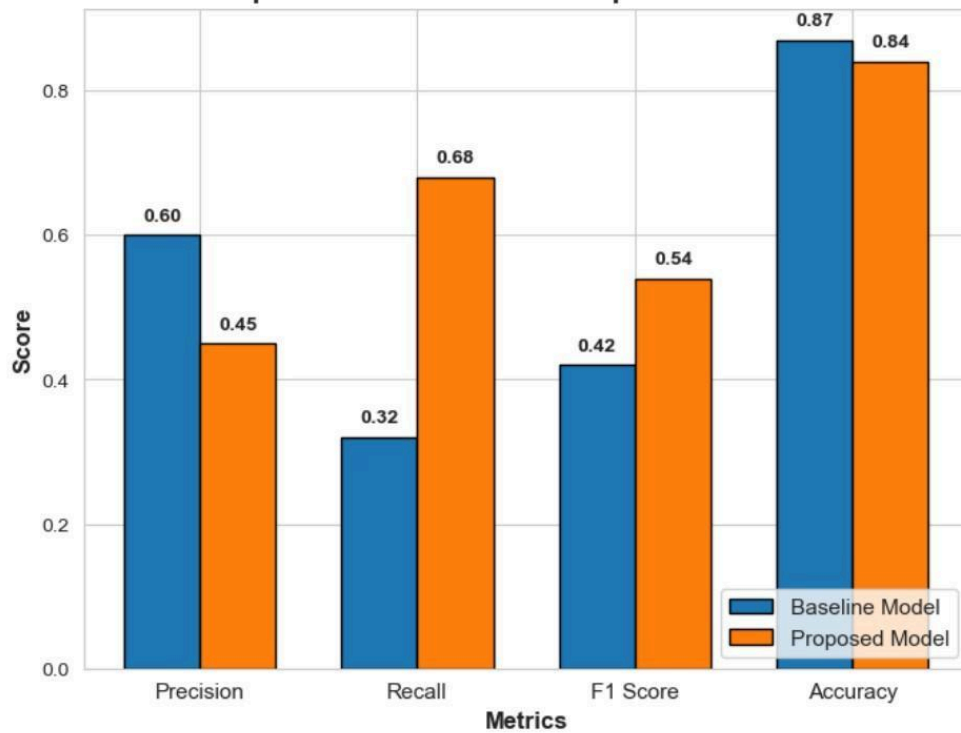


Figure 33: ADABOOST

Performance Comparison of GradientBoost: Proposed Model vs. Baseline Model

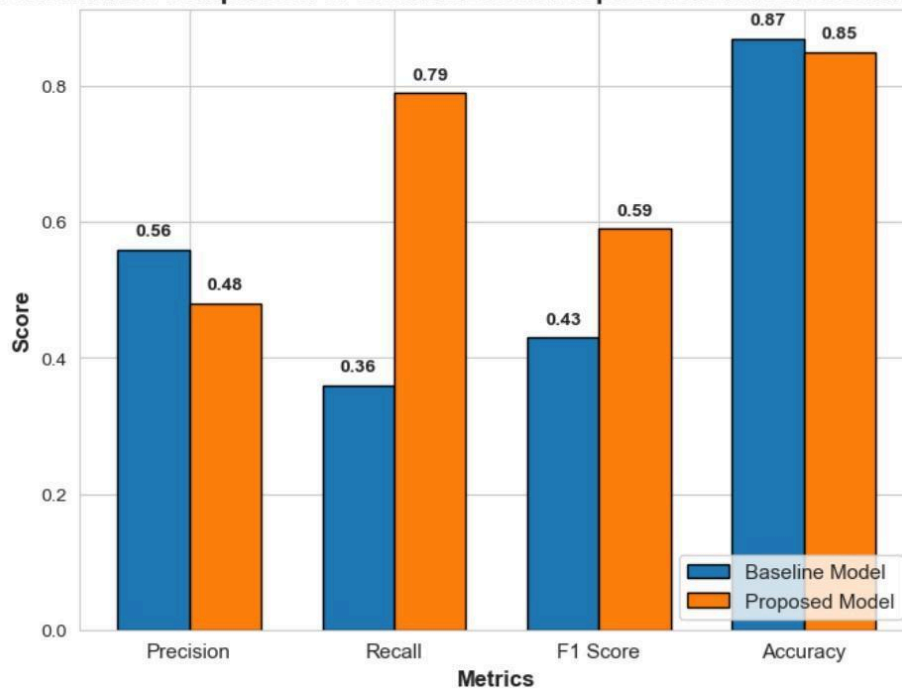


Figure 34: Gradient Boost

Performance Comparison of Catboost: Proposed Model vs. Baseline Model

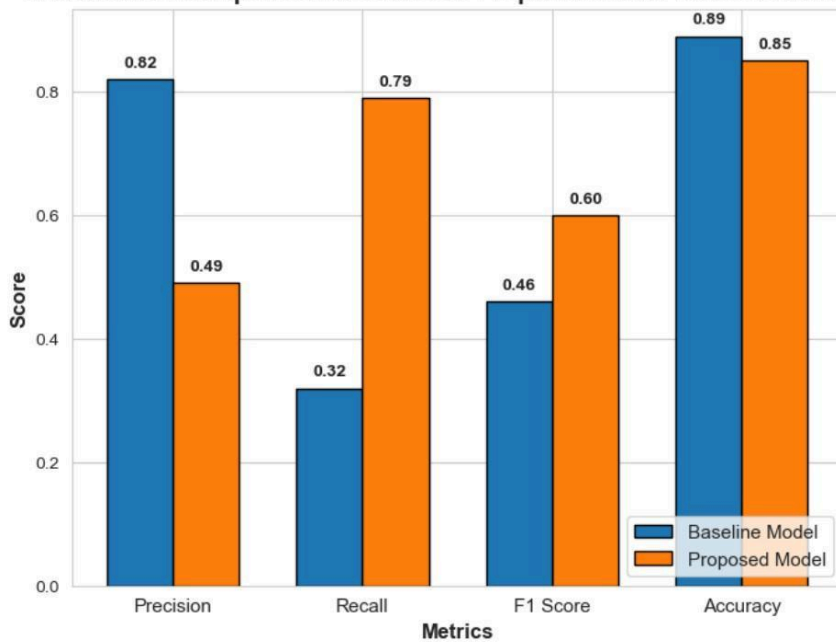


Figure 35: Catboost

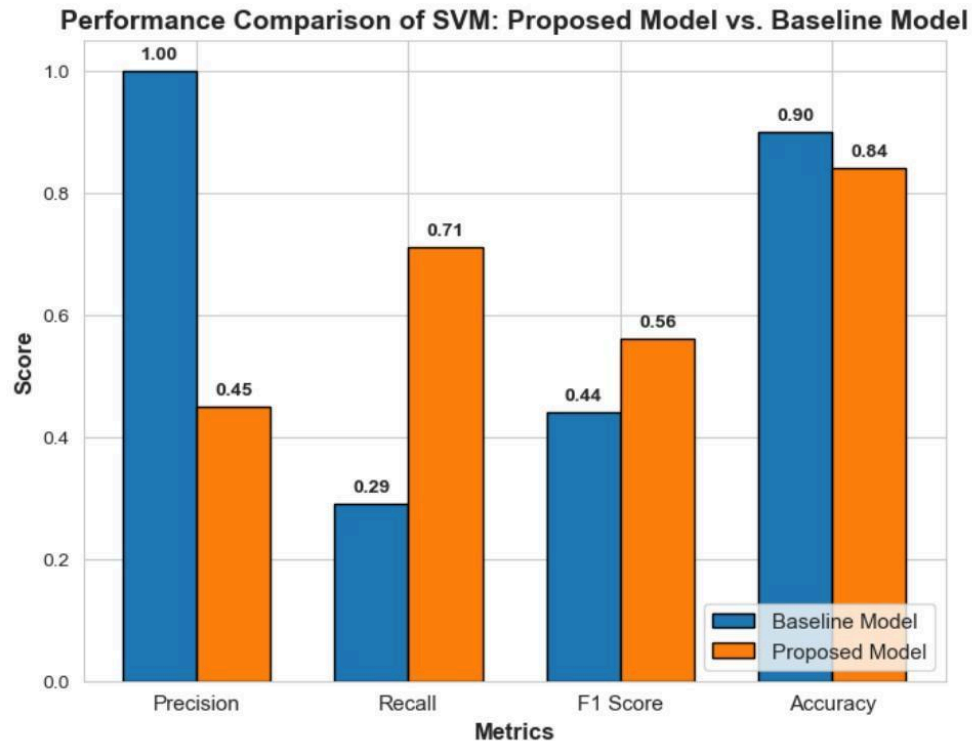


Figure 36: SVM

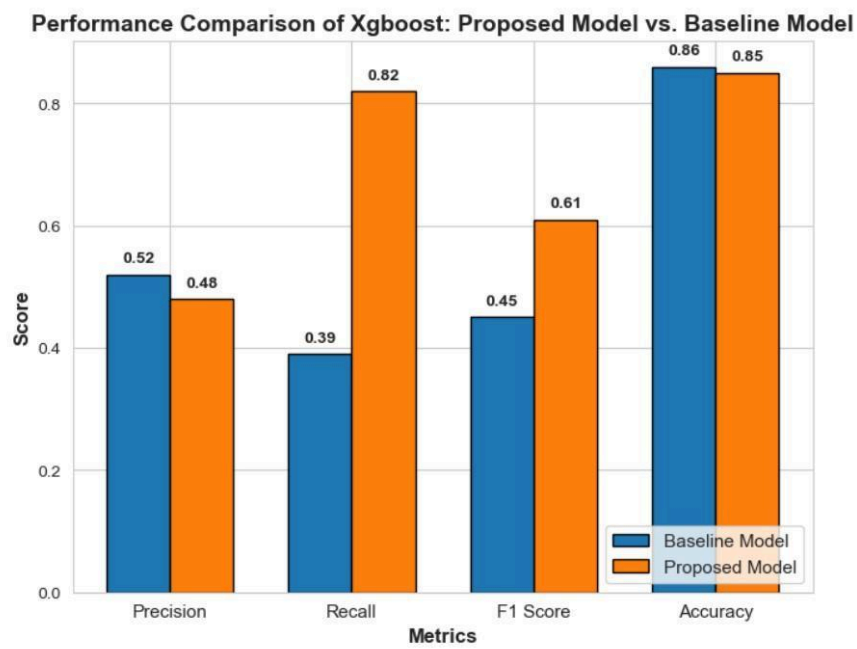


Figure 37: XGboost

The confusion matrices of each classifier are shown for better understanding in the following figures.

Logistic Regression:

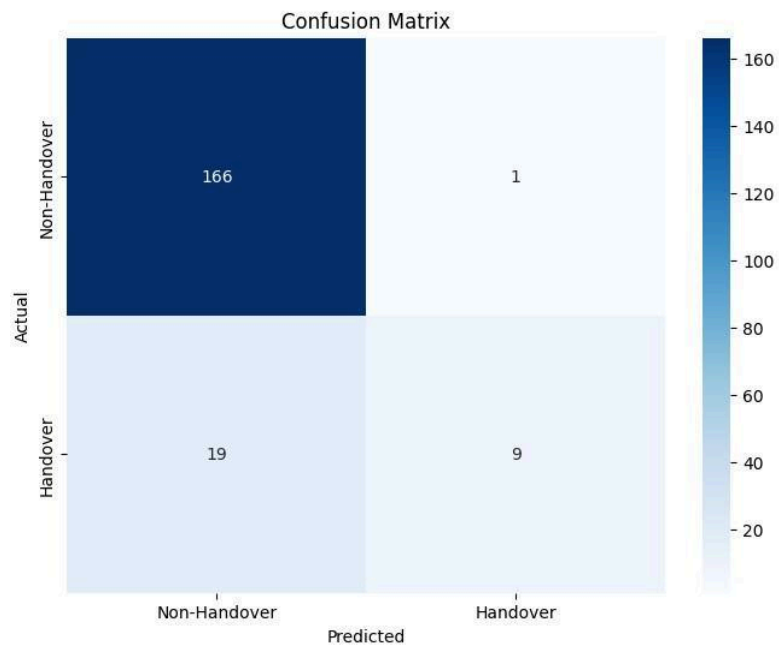


Figure 38: Baseline Model (Logistic Regression)

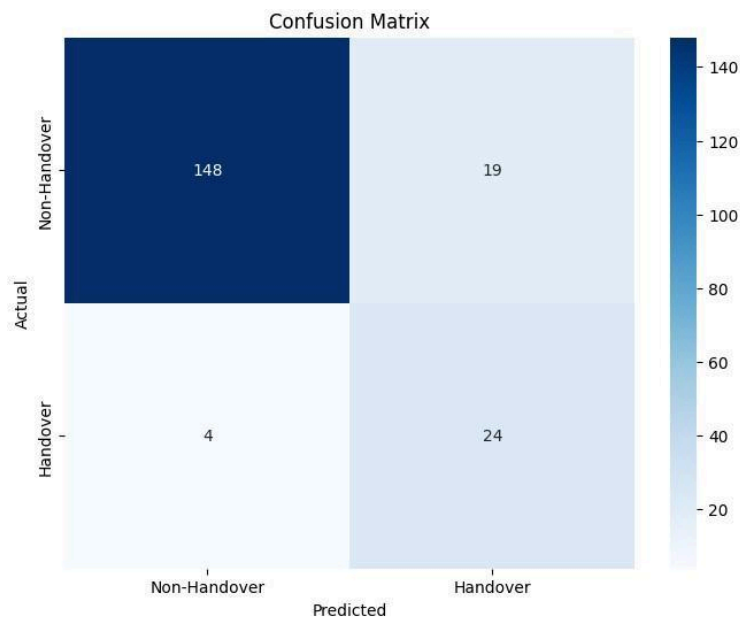


Figure 39: Proposed Model(Logistic Regression)

Decision Tree:

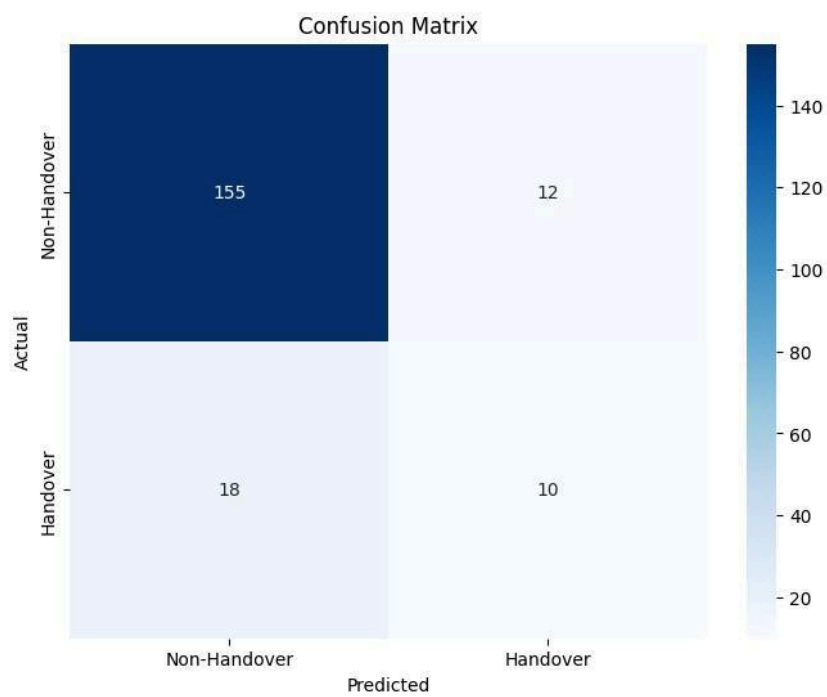


Figure 40: Baseline Model (Decision Tree)



Figure 41: Proposed Model(Decision Tree)

Random Forest:

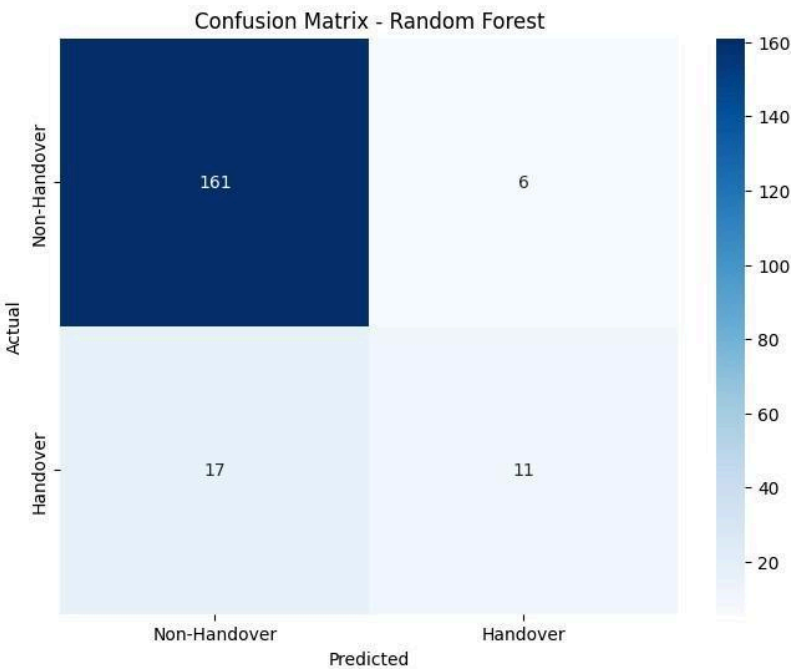


Figure 42: Baseline Model (Random Forest)

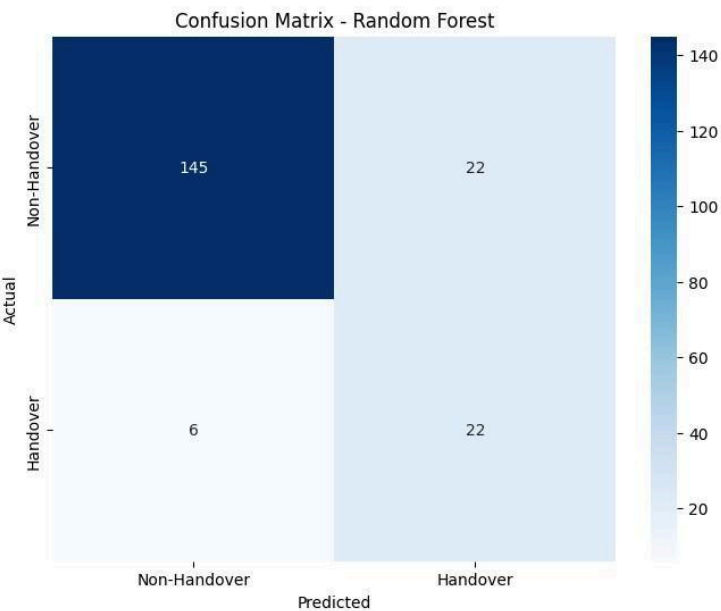


Figure 43: Proposed Model(Random Forest)

KNN:

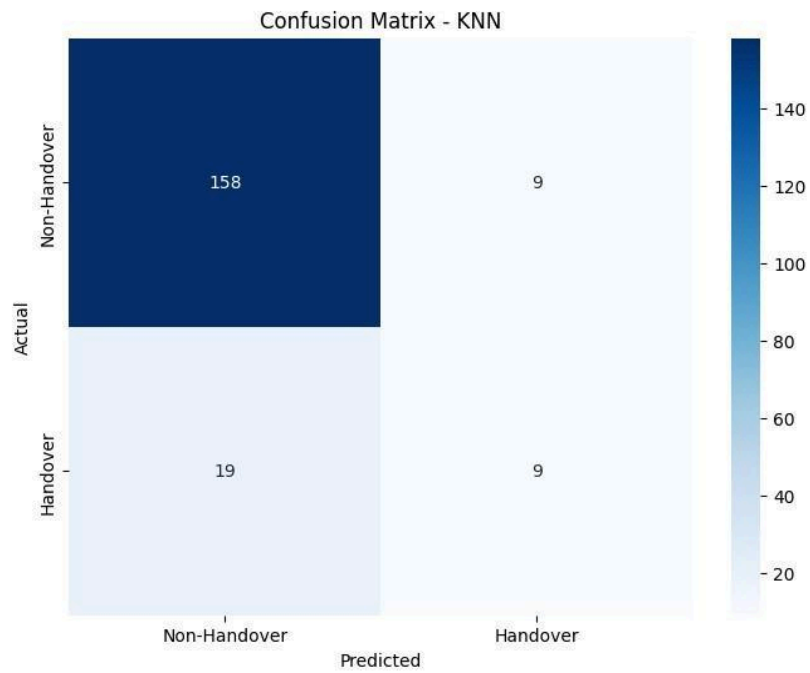


Figure 44: Baseline Model (KNN)

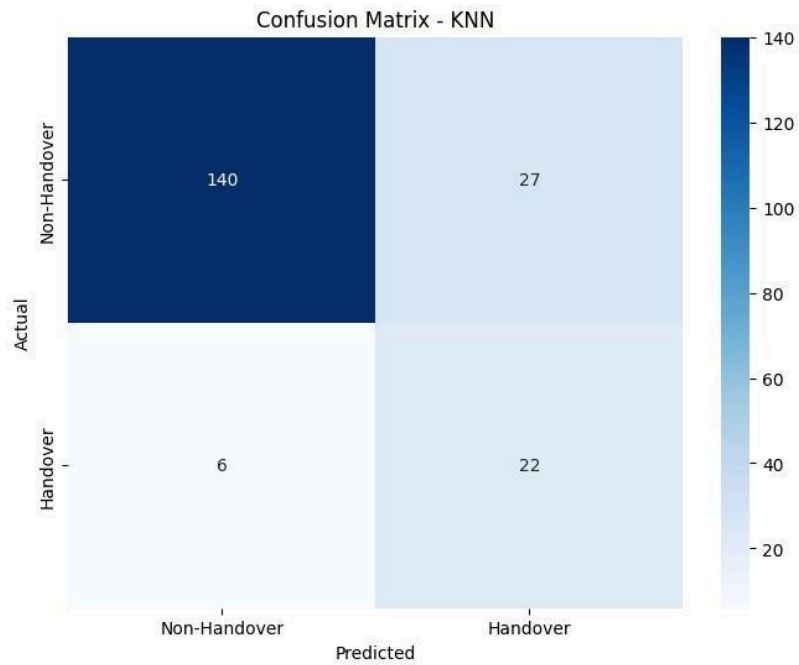


Figure 45: Proposed Model(KNN)

SVM:

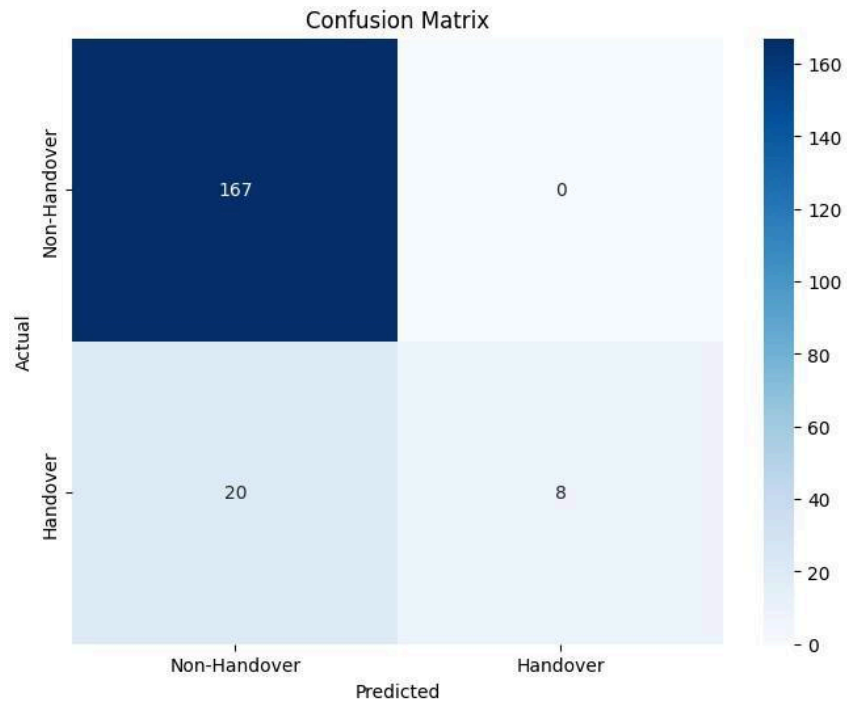


Figure 46: Baseline Model (SVM)

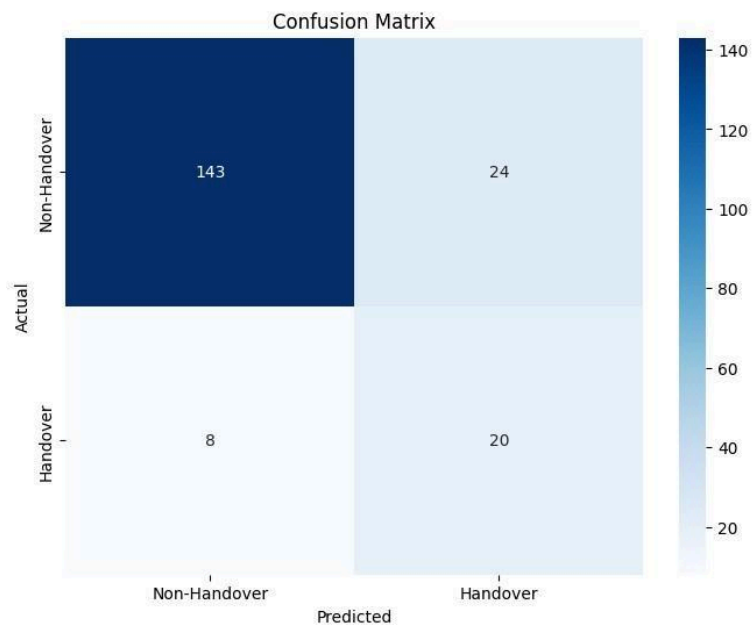


Figure 47: Proposed Model(SVM)

Naive Bayes:

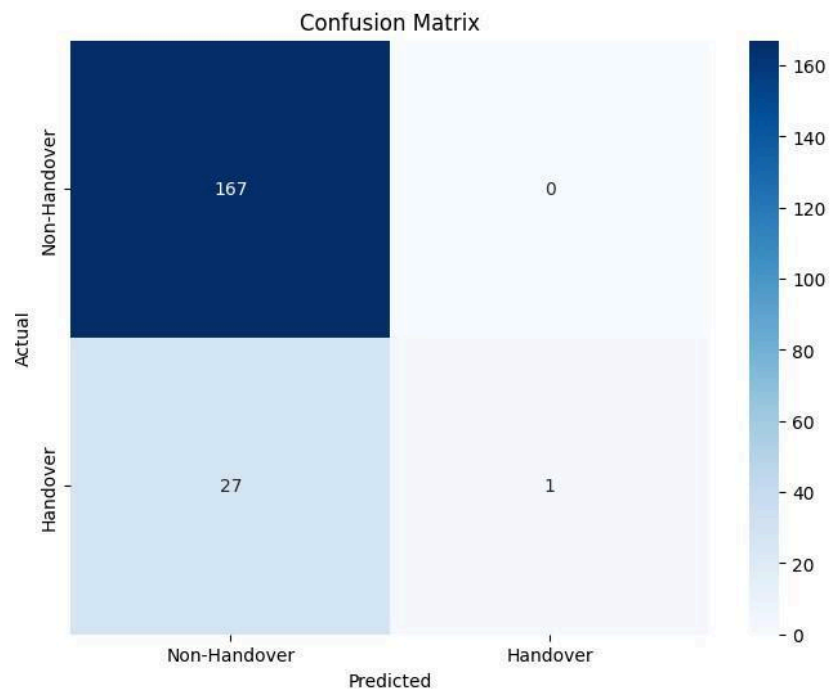


Figure 48: Baseline Model (Naive Bayes)

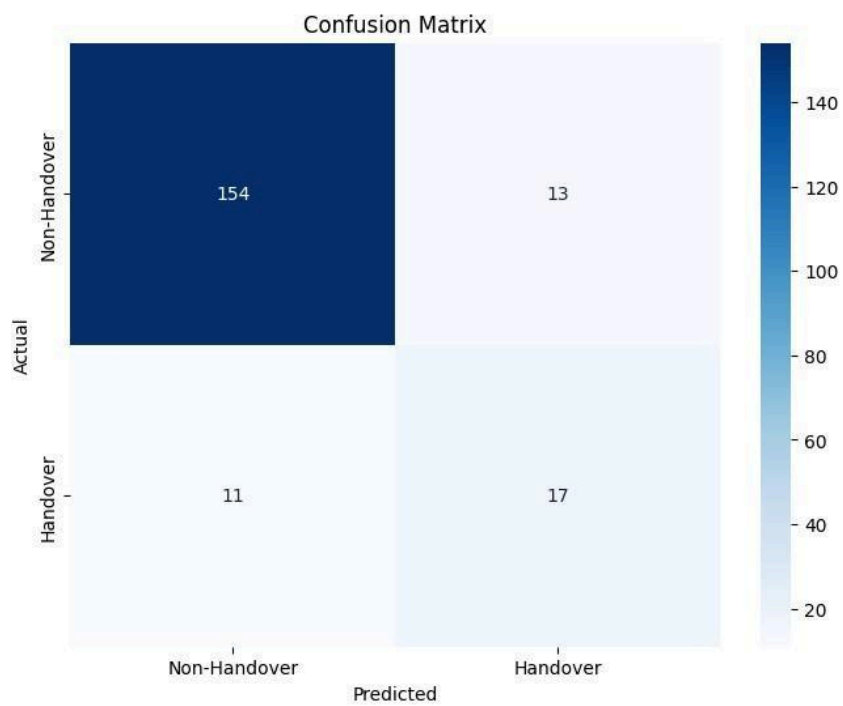


Figure 49: Proposed Model (Naive Bayes)

Adaboost:

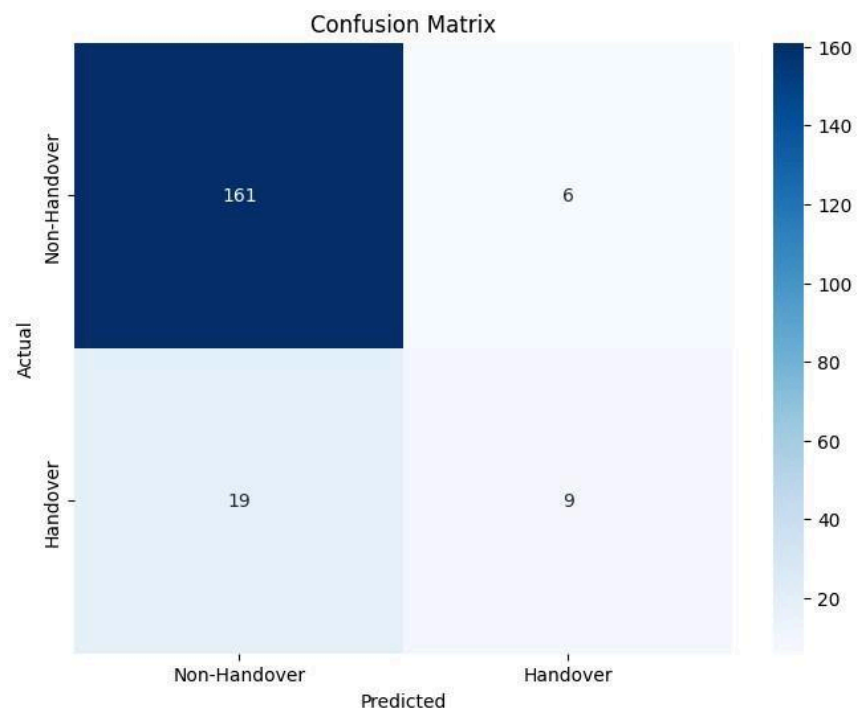


Figure 50: Baseline Model (Adaboost)

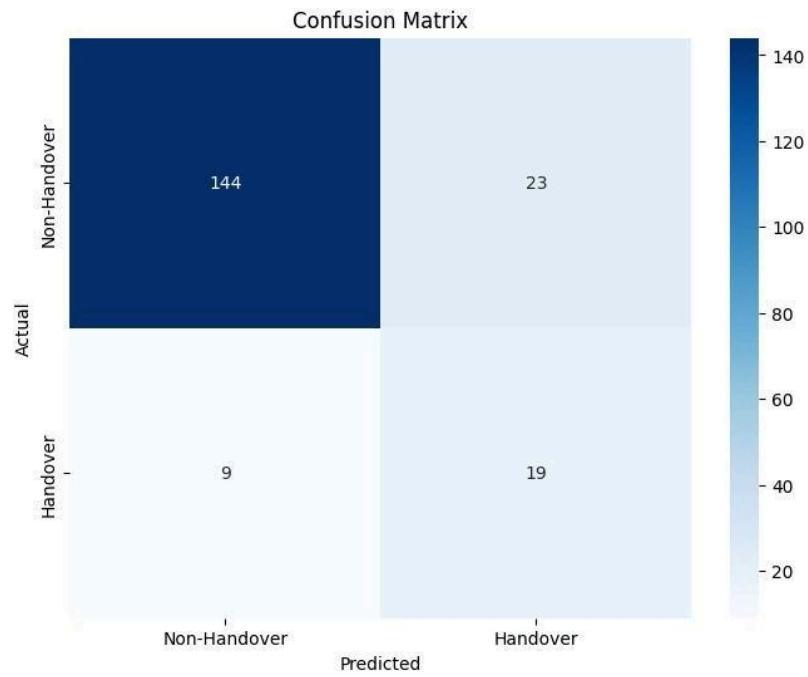


Figure 51: Proposed Model(Adaboost)

Catboost:

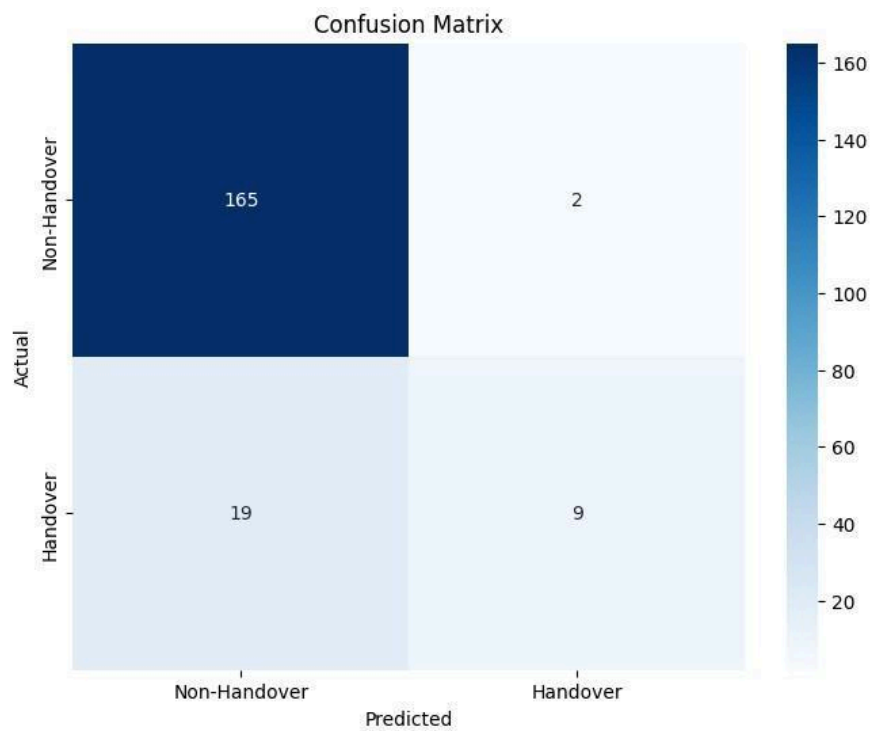


Figure 52: Baseline Model (Catboost)

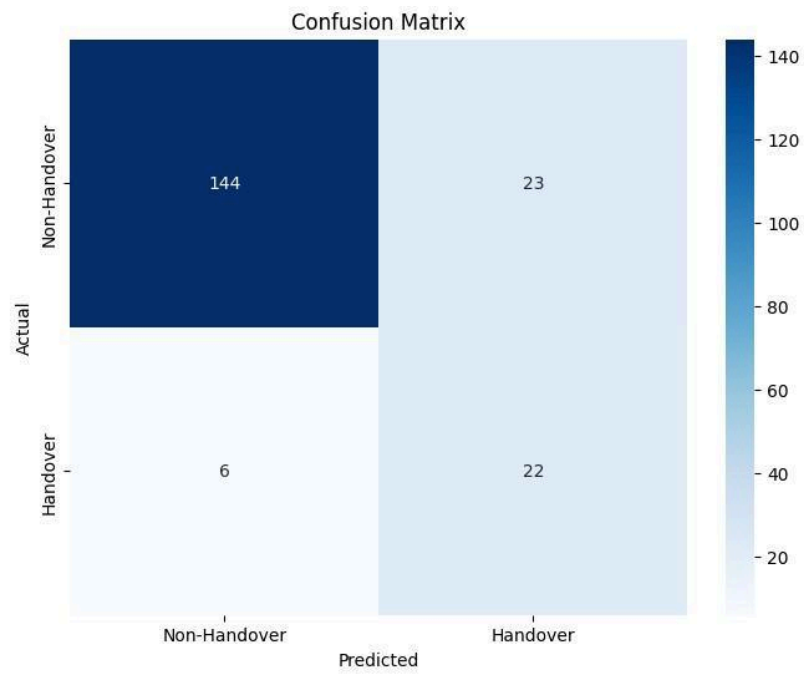


Figure 53: Proposed Model(Catboost)

Xgboost:

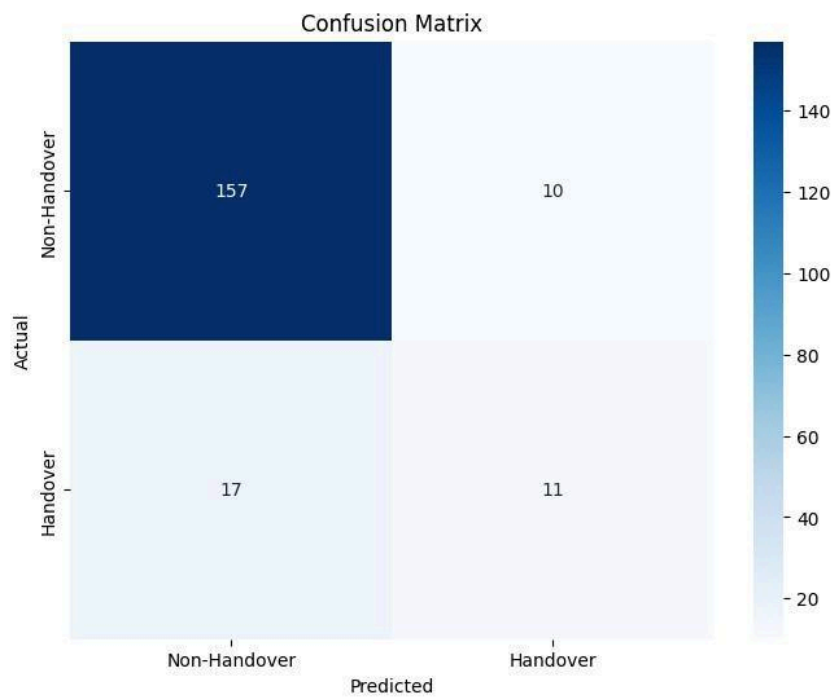


Figure 54: Baseline Model (Xgboost)

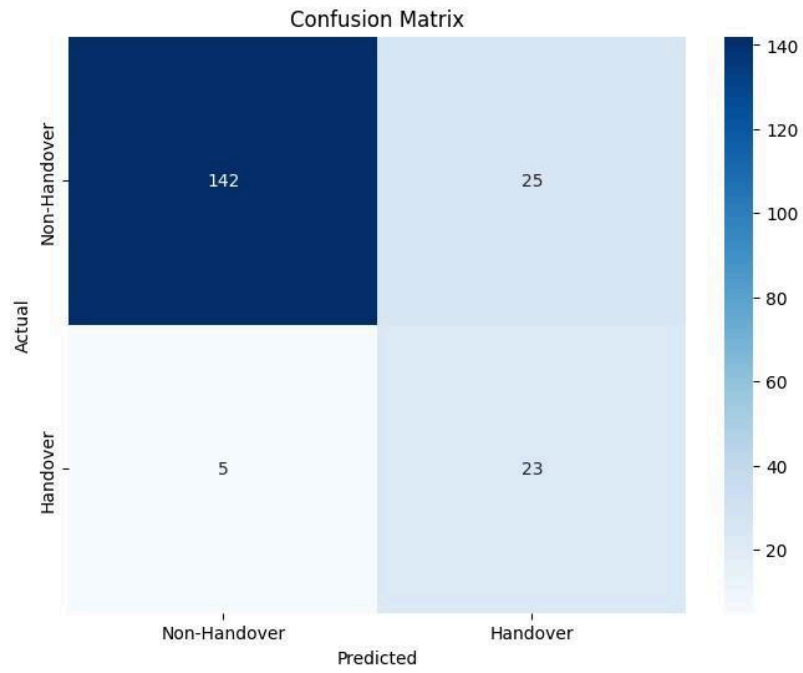


Figure 55: Proposed Model(Xgboost)

Gradientboost:



Figure 56: Baseline Model (Gradientboost)

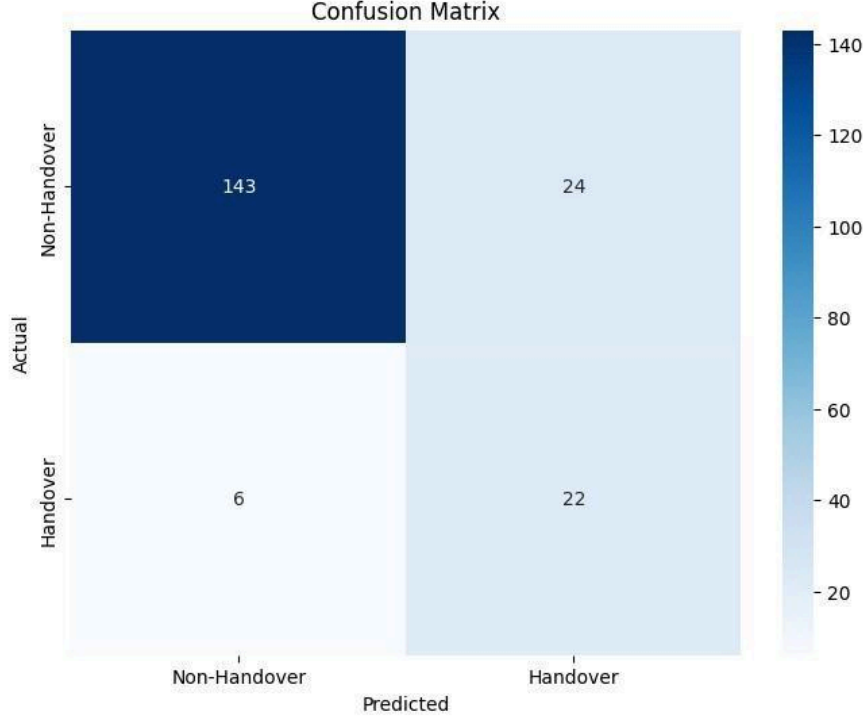


Figure 57: Proposed Model(Gradientboost)

The proposed model demonstrated a significant improvement over the baseline models in terms of Recall and F1-score, indicating a higher capability of capturing actual positive instances while maintaining a balanced trade-off between precision and recall. However, a slight reduction in Precision was observed, suggesting an increase in false positives. This trade-off was a deliberate consideration in model optimization, prioritizing higher recall to minimize false negatives, which is critical in our application domain. The rationale behind this decision and potential strategies to mitigate the precision drop will be discussed in detail in the subsequent section.

9.3 Experimental result evaluation of Handover Optimization:

As outlined in Chapter 7, our primary objective is to minimize unnecessary handovers to enhance the signal quality of a moving User Equipment (UE). To achieve this, we propose a hybrid learning framework that integrates supervised learning with Q-learning, leveraging a reward-based mechanism to optimize handover decisions. The model is designed to refine classifier-based predictions by strategically mitigating redundant handovers, ensuring that only necessary transitions occur between serving and neighboring cells.

The reward function plays a critical role in the reinforcement learning (RL) component of our model. It systematically penalizes handover decisions that do not yield improved signal quality while rewarding actions that maintain or enhance network performance. By incorporating parameters such as Reference Signal Received Power (RSRP), Reference

Signal Received Quality (RSRQ), and Carrier-to-Interference-plus-Noise Ratio (CINR) into the reward function, the Q-learning agent learns to differentiate between beneficial and detrimental handover events. This targeted reinforcement mechanism enables the model to dynamically adjust handover thresholds and prioritize network stability over frequent cell transitions.

In our dataset, we initially observed 94 handover instances across 647 data points. During the training phase, our model recorded 276 handover events, allowing the Q-learning agent to iteratively refine its decision-making policy. Subsequently, in the evaluation phase—where the trained Q-table was applied across the entire dataset—the number of handovers was reduced to 71 instances, indicating that the remaining handovers were identified as unnecessary and successfully suppressed. This reduction demonstrates the model’s effectiveness in preventing unnecessary handovers while preserving seamless connectivity.

Comparison of Handover Instances Before and After Hybrid Model

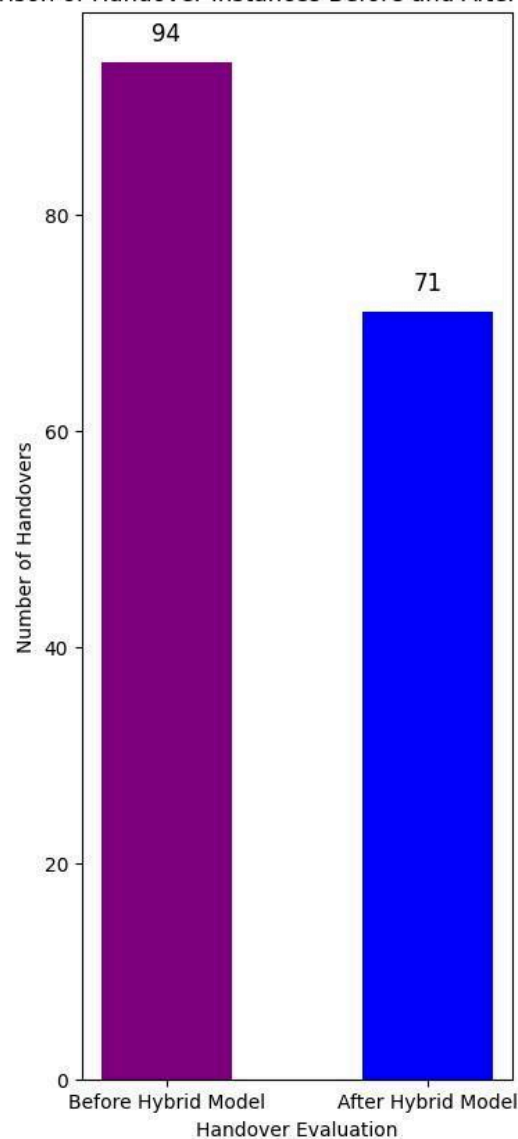


Figure 58: Number of Handover Before and After applying our model

Minimizing redundant handovers is crucial in wireless communication systems, particularly in LTE and 5G networks, where frequent transitions between cells can lead to increased signaling overhead, network congestion, and degraded Quality of Service (QoS). Excessive handovers also contribute to higher latency, packet loss, and energy consumption, negatively impacting both network efficiency and user experience. By intelligently regulating handover events, our proposed model enhances overall network performance, ensuring improved signal stability, lower handover failure rates, and optimized resource utilization. This adaptive framework can be further extended to next-generation cellular networks, facilitating intelligent mobility management in real-world deployments.

Chapter 9

Discussion

9.1 Critical Evaluation on QSI

In this study, we employed K-Means clustering to categorize Quality Signal Indicator (QSI) data, using manually specified initial centroids based on predefined QSI categories. Unlike supervised classification, where labels guide the learning process, K-Means operates unsupervised, relying solely on feature-space distances and iterative centroid optimization. While the clustering generally followed the predefined QSI categories, discrepancies arose due to the algorithm's independent reassignment of data points based on proximity to centroids. For instance, some data initially categorized as "Good" were reassigned to "Excellent" in K-Means, demonstrating its data-driven approach. To assess the alignment between the predefined categories and the K-Means results, we calculated category-wise percentage matches, providing insights into the effectiveness of centroid-based initialization in guiding the unsupervised learning process.

Table 11: Performance matching of supervised and unsupervised Model

Category	Percentage Matches
Excellent	76.92%
Good	65.52%
Moderate	71.43%
Poor	80.41%

9.2 Critical Evaluation on Handover Prediction Model

The underlying cost sensitivity connected to various classes is another element providing unbalanced data. Misclassifying one class can have more negative effects than misclassifying another in several situations. In credit card fraud detection, for instance, the financial institution and the client may suffer large losses in value if a fraudulent transaction is not detected. Misclassifying fraudulent activities therefore comes at a higher cost than misclassifying lawful transactions.

In the context of our study, the dataset exhibits a significant class imbalance, with a notably lower number of handover instances compared to non-handover instances. This imbalance poses a challenge for conventional classification metrics such as accuracy and precision, as these metrics may not provide a comprehensive assessment of model performance under such conditions. Specifically, a classifier that predicts the majority class (non-handover) more frequently can achieve a deceptively high accuracy, even if it fails to correctly identify actual handover events. Similarly, a high precision does not necessarily indicate a well-performing model in this scenario, as it focuses on minimizing false positives while potentially disregarding false negatives.

In our case, recall emerges as a more critical metric, as it quantifies the ability of our model to correctly identify actual handover instances (true positives) while minimizing false negatives. A false negative in this context—where an actual handover event is misclassified as a non-handover—can be costly, leading to degraded network performance, increased packet loss, and potential call drops. Given the high cost associated with undetected handovers, maximizing recall ensures that the model prioritizes detecting handover instances, even at the expense of some false positives.

Moreover, to balance precision and recall, we incorporate the F1-score, which provides a harmonic mean of both metrics. The F1-score is particularly useful in imbalanced classification problems, as it accounts for both false positives and false negatives, ensuring a more holistic evaluation. In our study, despite a slight reduction in precision, a significant improvement in recall and F1-score indicates that the model is effectively identifying handover instances while maintaining an overall balanced performance.

Considering the severe consequences of misclassifying handovers, a cost-sensitive approach is necessary, where false negatives (missed handovers) are penalized more heavily than false positives (unnecessary handovers). Our model is designed with this objective in mind, ensuring that recall remains the dominant metric, as detecting handovers at any cost is the priority to maintain optimal network quality and user experience.

9.3 Critical Evaluation on Handover Optimization

Our Hybrid Q-Learning Model is designed to optimize handover decisions in wireless networks by intelligently reducing unnecessary handovers while ensuring seamless connectivity. This is particularly useful in high-mobility scenarios such as urban cellular networks, where frequent handovers can degrade Quality of Service (QoS).

The model utilizes a user-defined reward function to guide learning, where penalties discourage poor handover decisions (e.g., switching to a weaker cell) and rewards incentivize optimal handovers (e.g., moving to a stronger signal cell with higher CINR, RSRP, and RSRQ). This function enables the model to balance network stability and mobility efficiency.

Additionally, the exploration-exploitation ratio plays a critical role in training. Initially, the model explores different handover decisions to learn their impact, and

as training progresses, it increasingly exploits the learned Q-values for optimal decisions. This ensures that the model generalizes well to real-world network variations, leading to adaptive, data-driven handover management that enhances overall network performance.

Chapter 10

Future Work

The current work successfully implements a hybrid Q-learning model for optimizing handover decisions in LTE networks. While it demonstrates a clear reduction in unnecessary handovers and improved decision-making, several areas remain for further exploration and development. This chapter outlines the key directions that can enhance and extend the scope and performance of the current system.

10.1 Integration of Deep Q-Networks (DQN)

In the next phase of this research, we plan to replace the tabular Q-learning algorithm with Deep Q-Networks (DQN). Unlike tabular Q-learning, DQN utilizes a neural network to approximate the Q-function, allowing the agent to handle large and continuous state spaces more efficiently. This is particularly advantageous in mobile network environments, where signal conditions and user behavior can vary dynamically over time and location. By employing DQN, the model will be capable of generalizing better from unseen states and learning more robust handover policies under diverse mobility conditions.

10.2 Policy-Based Reinforcement Learning

Another major area of future work involves the application of policy-based reinforcement learning, specifically Policy Gradient algorithms. Unlike value-based methods like Q-learning and DQN, policy gradient methods directly learn the optimal policy by optimizing a performance objective. This allows for more flexible action selection and can be especially useful in environments where discrete actions (e.g., handover vs. no handover) need to be fine-tuned over time.

We plan to experiment with algorithms such as REINFORCE and Actor-Critic methods to evaluate their effectiveness in handover optimization. These methods have the potential to improve convergence speed and policy smoothness, which are essential in real-time decision-making scenarios.

10.3 Multi-Objective Optimization

The current model focuses primarily on reducing handover frequency. However, in real-world network scenarios, handover decisions affect multiple Key Performance Indicators (KPIs), such as latency, throughput, and packet loss. A future extension of this work will involve multi-objective optimization, where the reinforcement learning agent will learn to balance various performance metrics to achieve a more holistic QoS improvement.

10.4 Real-Time Implementation and 5G/LEO Extension

Further work can also be directed toward real-time implementation of the optimized handover model using edge computing or mobile network APIs. This would involve integrating the ML/RL model into live testbeds or simulation platforms for LTE/5G/LEO satellite environments. Doing so will validate the model's practical viability and adaptability under operational conditions, including high-speed mobility and dense urban coverage.

Chapter 11

Demonstration of Outcome Based Education (OBE)

11.1 Course Outcomes (COs) Addressed

The following table shows the COs addressed in EEE 4700 for Project and Thesis.

COs	CO Statement	POs	Put Tick (✓)
			EEE 4700
CO1	Identify a contemporary real life problem related to electrical and electronic engineering by reviewing and analyzing existing research works.	PO2	✓
CO2	Determine functional requirements of the problem considering feasibility and efficiency through analysis and synthesis of information.	PO4	✓
CO3	Select a suitable solution and determine its method considering professional ethics, codes and standards.	PO8	✓
CO4	Adopt modern engineering resources and tools for the solution of the problem.	PO5	✓
CO5	Prepare management plan and budgetary implications for the solution of the problem.	PO11	✓

CO6	Analyze the impact of the proposed solution on health, safety, culture and society.	PO6	√
CO7	Analyze the impact of the proposed solution on environment and sustainability.	PO7	√
CO8	Develop a viable solution considering health, safety, cultural, societal and environmental aspects.	PO3	
CO9	Work effectively as an individual and as a team member for the accomplishment of the solution.	PO9	√
CO10	Prepare various technical reports, design documentation, and deliver effective presentations for demonstration of the solution.	PO10	√
CO11	Recognize the need for continuing education and participation in professional societies and meetings.	PO12	

11.2 Aspects of Program Outcomes (POs) Addressed

The following table shows the aspects addressed for certain Program Outcomes (POs) addressed in EEE 4700 for Project and Thesis.

	Statement	Different Aspects	Put Tick (√)
PO3	Design/development of solutions: Design solutions for complex electrical and electronic engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	Public health	
		Safety	
		Cultural	
		Societal	√
		Environmental	
PO4	Investigation: Conduct investigations of complex electrical and electronic engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Design of experiments	√
		Analysis and interpretation of data	√
		Synthesis of information	√
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex electrical and electronic engineering problems.	Societal	√
		Health	
		Safety	
		Legal	√
		Cultural	
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts.	Societal	√
		Environmental	

PO8	Ethics: Apply ethical principles embedded with religious values, professional ethics and responsibilities, and norms of electrical and electronic engineering practice.	Religious values	
		Professional ethics and responsibilities	√
		Norms	√
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Comprehend and write effective reports	√
		Design documentation	√
		Make effective presentations	√
		Give and receive clear instructions	√
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	Engineering management principles	√
		Economic decision-making	√
		Manage projects	√
		Multidisciplinary environments	

11.3 Knowledge Profiles (K3 – K8) Addressed

The following table shows the Knowledge Profiles (K3 – K8) addressed in EEE 4700 for Project and Thesis.

K	Knowledge Profile (Attribute)	Put Tick (√)
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	√
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	√
K5	Knowledge that supports engineering design in a practice area	√
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	√
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability	
K8	Engagement with selected knowledge in the research literature of the discipline	√

The following table explains or justifies how the Knowledge Profiles (K3 – K8) have been addressed in EEE 4700/4800 (Project and Thesis).

K	Explanation/Justification
K3	The systematic, theory-based formulation of engineering fundamentals essential for the engineering discipline directly supports our thesis on handover optimization by providing a solid foundation in telecommunications principles, signal processing, and network design. Understanding these core theories enables the development of advanced algorithms and models that accurately predict and optimize handover processes, ensuring improved network performance and enhanced user experience. By applying these engineering fundamentals, the project aims to address real-world challenges in mobile communications effectively, contributing to the overall advancement of the telecommunications field.
K4	The thesis on handover optimization leverages specialized knowledge in telecommunications engineering, particularly in areas like radio frequency (RF) management, mobility protocols, and machine learning. These theoretical frameworks form the foundation for developing innovative models that enhance handover decision-making processes. By using cutting-edge techniques, the research remains at the forefront of the discipline, advancing best practices in optimizing network performance and reliability in mobile communication systems
K5	In the context of handover optimization, the knowledge supporting engineering design includes understanding signal strength metrics (RSRP, RSSI, RSRQ), mobility management, and machine learning techniques. This foundational knowledge guides the design of predictive models that enhance handover decisions in real-time, ensuring efficient network performance. By applying this knowledge, the design of the optimization process reduces unnecessary handovers and improves user experience, aligning with best practices in mobile network engineering.
K6	In handover optimization, engineering practice knowledge involves the application of technologies such as XCAL for data collection during drive tests and machine learning for predictive analysis. This knowledge includes practical skills in configuring and managing network parameters (e.g., signal strength, TTT, and cell handover events), and leveraging tools for real-time network monitoring and optimization. It ensures the effective implementation of handover solutions in live networks, aligning with industry standards and practices in telecommunications engineering.
K7	
K8	In the context of handover optimization using machine learning, engagement with the research literature involves critically analyzing and applying relevant studies on signal strength metrics, mobility management, and predictive models for handover decisions. By exploring existing frameworks and methodologies, such as machine learning-based optimization techniques, the research leverages state-of-the-art findings to improve network efficiency and user experience. This engagement also includes evaluating advancements in mobility management, network performance, and the use of data-driven approaches to enhance handover timing and reduce unnecessary transitions in mobile networks.

Use of Complex Engineering Problems

Handover optimization in mobile communication networks presents a complex engineering challenge due to the dynamic nature of network conditions, user mobility, and varying signal quality. To address this complexity, a systematic approach involving data-driven methodologies is employed. By collecting real-time data during drive tests using XCAL, we gather comprehensive information on signal strength metrics, user equipment speed, and historical handover events. Analyzing this data through advanced analytics allows us to identify patterns and correlations, which serve as the foundation for predictive modeling. This enables us to anticipate the optimal timing and conditions for handovers, minimizing unnecessary transitions that could disrupt user experience.

Moreover, incorporating machine learning techniques provides a robust framework for decision-making in handover management. Through the training of various machine learning models, we can evaluate their performance in predicting when handovers are necessary. By comparing models based on their accuracy, precision, and recall, we identify the best fit for our specific scenario. This iterative process of model selection and refinement allows us to enhance the efficiency of handover decisions, ultimately resulting in a reduction in failed handovers and improved overall network performance.

Lastly, the importance of stakeholder engagement cannot be overlooked in the optimization process. Collaboration with network operators and engineers ensures that the developed solutions align with practical deployment scenarios and operational constraints. Continuous feedback loops from real-world implementations facilitate the iterative improvement of our models and strategies, allowing us to adapt to changing network conditions and user behaviors. This integrated approach not only addresses the technical challenges of handover optimization but also ensures that solutions are viable and accepted within the operational framework of mobile networks.

Socio-Cultural, Environmental, And Ethical Impact

Socio-Cultural Impact

The handover optimization project has the potential to significantly enhance the user experience in mobile communications, which is integral to modern social interactions and cultural engagement. By reducing call drop rates and improving overall connectivity, the project supports seamless communication for individuals and communities, fostering stronger social ties and greater access to information. Enhanced mobile services can empower users in remote areas, bridging the digital divide and facilitating social inclusion. Moreover, improved connectivity can bolster local economies by supporting businesses that rely on mobile communication, enabling them to reach a broader audience and engage in e-commerce.

Environmental Impact

From an environmental perspective, optimizing handover processes can lead to reduced energy consumption in mobile networks. Fewer unnecessary handovers mean that user equipment (UE) spends less time engaging in signaling processes, leading to lower power usage. This not only benefits individual users through longer battery life but also contributes to the overall energy efficiency of network infrastructure. By minimizing the signaling load on base stations, the project can potentially reduce carbon emissions associated with energy consumption in mobile networks. Ultimately, a more efficient network can support sustainability initiatives by promoting eco-friendly practices in the telecommunications sector.

Ethical Impact

The ethical considerations of the handover optimization project revolve around data privacy and algorithmic fairness. As the project involves the collection of real-time user data through drive tests, it is essential to ensure that data handling practices adhere to privacy regulations and ethical standards. Transparency in data usage and obtaining informed consent from users will be prioritized to maintain trust. Additionally, machine learning models used for predicting handovers must be designed to avoid bias, ensuring that all users receive equitable service regardless of their location or usage patterns. Addressing these ethical concerns will not only enhance the project's credibility but also foster responsible innovation in the field of telecommunications.

11.4 Attributes of Ranges of Complex Engineering Problem Solving (P1 – P7) Addressed

The following table shows the attributes of ranges of Complex Engineering Problem Solving (P1 – P7) addressed in EEE 4700 for Project and Thesis.

P	Range of Complex Engineering Problem Solving	Put Tick (✓)
Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:	
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach	
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues	✓
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	✓
Familiarity of issues	P4: Involve infrequently encountered issues	✓
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering	
Extent of stakeholder involvement and	P6: Involve diverse groups of stakeholders with widely varying needs	

conflicting requirements		
Interdependence	P7: Are high level problems including many component parts or sub-problems	√

The following table explains or justifies how the attributes of ranges of Complex Engineering Problem Solving (P1 – P7) have been addressed in EEE 4700/4800 (Project and Thesis).

P	Explanation/Justification
P1	Handover optimization in mobile networks requires in-depth engineering knowledge, particularly in wireless communication principles (K3), signal processing (K4), and predictive modeling (K6). The challenge involves applying first principles and fundamentals, such as radio signal propagation and machine learning algorithms, to resolve complex, data-driven decisions for optimal handover timing.
P2	Conflicting requirements arise in balancing user experience, network performance, and energy efficiency. For example, reducing handovers can conserve resources but might degrade service quality if poorly timed. The optimization needs to address both technical performance and user satisfaction, making it a multifaceted issue.
P3	The problem lacks an obvious solution due to the dynamic nature of mobile networks, which requires abstract thinking and modeling. Creating predictive models for handover decisions demands originality, as real-time factors like user mobility, signal fluctuation, and network congestion need to be accounted for.
P4	Handover optimization encounters infrequent issues, especially with the emergence of new technologies like LTE, 5G. Problems such as ping-pong handovers or rare network outages require tailored solutions, often necessitating innovative analysis beyond standard procedures.
P5	
P6	
P7	Handover optimization is a high-level engineering challenge that consists of multiple sub-problems, including signal strength assessment, mobility tracking, and real-time decision-making. Each sub-problem is interdependent, requiring holistic consideration of various factors to ensure network efficiency and user satisfaction.

11.5 Attributes of Ranges of Complex Engineering Activities (A1 – A5)

Addressed

The following table shows the attributes of ranges of Complex Engineering Activities (A1 – A5) addressed in EEE 4700 for Project and Thesis.

A	Range of Complex Engineering Activities	Put Tick
Attribute	Complex activities means (engineering) activities or projects that have some or all of the following characteristics:	(√)
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)	√

Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues	√
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways	√
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation	
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches	√

The following table explains or justifies how the attributes of ranges of Complex Engineering Activities (A1 – A5) have been addressed in EEE 4700/4800 (Project and Thesis).

A	Explanation/Justification
A1	The thesis project on handover optimization utilizes diverse resources such as data collection tools (XCAL), machine learning technologies, financial budgeting for drive tests, and collaboration with network engineers. It also involves managing large-scale datasets and computational resources for model training and evaluation.
A2	The project addresses significant problems, such as conflicting requirements between minimizing handovers and maintaining service quality. These technical challenges require balancing network performance, user experience, and operational efficiency, making it a multi-dimensional task with intricate interactions.
A3	The project applies machine learning in novel ways to optimize handover processes, which is a creative use of engineering principles. By integrating real-time network data and predictive algorithms, the research develops innovative solutions beyond conventional handover strategies used in mobile networks.
A4	
A5	This project extends beyond previous experiences by applying principles of signal processing and machine learning to the evolving challenges of mobile networks. The methods used build upon established concepts but adapt them to new, real-world challenges in 4G and beyond.

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